

M&A OPERATING SYSTEM

Data Design Document

CPG Manufacturing Operations — Data Design Document

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About This Document

Supplied text. Held in `boilerplate.json`, which this run does not carry.

Version

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1 Introduction

1.1 About M&A Operating System

Supplied text. Held in `boilerplate.json`, which this run does not carry.

1.2 About the MAOS Practical Data Modelling Approach

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1.3 Summary of Findings

The model covers the making of consumer packaged goods across five subgroups, in the organisation's own factories and at outside manufacturers. It reuses five DDA domains and proposes five more, and the chain of what goes into what stops at anything bought in, where the supplier's declaration takes over.

	Count
Business requirements	69
Data requirements	36
Subject domains	5 in DDA · 5 proposed
Roles	70
Relationships	38
Concepts	66
Attributes	120
Lookups	9
Business test cases	38 pass · 3 with a limit or a gap
Sources	40

Status: **published with accepted limits**, as a reference model.

2 Scope

2.1 The business

A manufacturer of consumer packaged goods. It buys materials, makes and tests product, and releases it for despatch, through its own factories and through outside manufacturers.

Subgroups covered, equally: Food and beverage, Household and cleaning products, Personal care and cosmetics, Pet care, Beauty.

2.2 In scope

ID	In scope
SCI001	Buying materials, and deciding which suppliers may supply them
SCI002	Defining products — what they are made from and what they must be
SCI003	Defining how products are made — the steps, the settings and the recipes
SCI004	The layout of each factory and the machines in it
SCI005	Making product, and what each batch actually used and produced
SCI006	Testing, recording problems, putting them right, and releasing product
SCI007	Tracing any batch back to its ingredients and forward to where it went
SCI008	Stock held at the organisation's own sites
SCI009	Product made or packed by outside manufacturers

2.3 Out of scope

ID	Out of scope	Belongs to
SCO001	Trade spending, promotions and pricing with retailers	The commercial model
SCO002	What happens on the shelf in store	The commercial model
SCO003	Market share and sales data bought from research firms	The commercial model
SCO004	Marketing, campaigns and advertising	The digital marketing model
SCO005	Accounts, ledgers and statutory reporting	The finance model
SCO006	Research and development before a product is formally defined	Not modelled

2.4 Shared with other models

ID	Object	In DDA
SCS001	People	DMD000000001

ID	Object	In DDA
SCS002	Legal entities	DMD000000012
SCS003	Documents	DMD000000016
SCS004	Events	DMD000000020
SCS005	Transactions	DMD000000022

2.5 Assumptions

ID	Assumption
SCA001	The factory, laboratory, quality and maintenance systems already in place stay in place. This model holds what they record; it does not replace them.
SCA002	Material held by a supplier or outside manufacturer is theirs. It becomes the organisation's on receipt, and the organisation holds no stock at any party it does not operate.
SCA003	Something bought as an ingredient or part may be another company's finished product. What it is made from is theirs and may not be known here.
SCA004	This is research-only work. No business owner has reviewed the requirements.

2.6 Open questions

ID	Question
SCQ001	Is finished output identified individually, or only by batch? The model supports both; how much of Instances is used depends on the answer.
SCQ002	What proportion of output is made by outside manufacturers? Where it is most of it, the organisation holds little production record of its own.
SCQ003	State transitions could not be read from the platform, so the change package carries no preconditions.

3 Business Requirements

Each states something true of the business, in words a business owner can agree or disagree with. The cases that show each one is met are in Appendix E, sorted by the same requirement ID.

ID	Category	Business requirement	Owner	Subgroup	Sources
REQ070	Cost	What a product costs to make is built up from what went into it and what was done to it, and expected and actual cost differ.	Finance lead		SRC014 · SRC018
REQ033	Equipment	A plant is organised into areas, lines and units, and work happens at each level.	Plant manager		SRC001 · SRC004
REQ034	Equipment	A place on a line and the machine standing in it are different things, and machines are replaced.	Maintenance lead		SRC001 · SRC004
REQ035	Equipment	A machine has a repair history, a record of how it fails and a working life that follow it wherever it goes.	Maintenance lead		SRC018 · SRC016
REQ036	Equipment	Some equipment matters more than others — because it stops the line, costs more to repair, or its failure is a safety issue.	Maintenance lead		SRC018
REQ037	Equipment	A stoppage is recorded by production as lost output and by maintenance as work, and both must agree it was one stoppage.	Plant manager		SRC016
REQ038	Equipment	Why a line stopped matters as much as that it stopped, and the reasons must be recorded consistently.	Plant manager		SRC017 · SRC018
REQ039	Equipment	Maintenance is planned, prompted by a condition, or done after a failure, and these are different kinds of work.	Maintenance lead		SRC016 · SRC018
REQ068	Identity	Two records may turn out to describe the same material, supplier or place.	Data steward		SRC005
REQ069	Identity	An identifier issued by an outside authority is the strongest evidence of what something is.	Data steward		SRC030 · SRC011
REQ001	Material	Everything the organisation buys, makes or sells is something it holds, counts and can run out of — ingredients, packaging, part-made goods and finished goods.	Supply chain lead		SRC005 · SRC039
REQ002	Material	A product is made from a defined set of inputs in defined quantities, and that definition changes over time.	Technical lead		SRC002 · SRC005 · SRC008
REQ003	Material	An input may itself be made from other inputs, to any depth, before anything is finished.	Technical lead		SRC005 · SRC006

ID	Category	Business requirement	Owner	Subgroup	Sources
REQ004	Material	Using an input means using all of it as it is defined. What goes into an input cannot be adjusted by whoever uses it; needing it different means needing a different input.	Technical lead		SRC005 · SRC030
REQ005	Material	The same input goes into many different products, and a product may be sold as it is and also used inside another.	Technical lead		SRC005 · SRC006
REQ006	Material	What a recipe says should go in and what a batch actually used differ, and the difference matters.	Plant manager		SRC005 · SRC033
REQ007	Material	An approved alternative may be used in place of a specified input, and which one was used on a batch must be known.	Technical lead		SRC005 · SRC033
REQ008	Material	A material may be bought from several approved suppliers, and approval may lapse or be suspended.	Procurement lead		SRC020 · SRC029
REQ009	Material	A material arrives with the supplier's statement of what it is, and the organisation may or may not check it.	Quality manager		SRC020 · SRC039
REQ010	Material	Something bought as an ingredient or part may be another company's finished product, and what it is made from may not be known.	Technical lead		SRC023 · SRC024
REQ011	Material	Properties of an input carry through to what is made from it — allergens and other hazards, certifications, origin, and the date by which it must be used.	Quality manager		SRC030 · SRC035 · SRC007
REQ012	Material	Food allergens must be declared on a product by the food they come from, including those that arrive inside another ingredient.	Quality manager	Food and beverage	SRC034 · SRC035 · SRC008
REQ013	Material	Household cleaning products must disclose each ingredient and how much of it there is, and some may be withheld as confidential.	Regulatory lead	Household and cleaning products	SRC025 · SRC026
REQ014	Material	The hazard rules a product must follow depend on what kind of product it is; food allergen rules do not apply to pet food, cosmetics or household cleaners.	Regulatory lead		SRC036 · SRC019 · SRC025 · SRC028
REQ015	Material	The shortest-lived input limits how long the finished product lasts.	Quality manager		SRC007 · SRC039
REQ016	Material	A batch may be split or combined for handling, and every part stays traceable to where it came from.	Supply chain lead		SRC005 · SRC007

ID	Category	Business requirement	Owner	Subgroup	Sources
REQ017	Material	Ingredients, part-made goods and finished goods are each held at places, and each is counted separately.	Supply chain lead		SRC039 · SRC005
REQ018	Material	Stock at one place may be available, on hold or rejected, and these are separate quantities rather than one quantity with a label.	Quality manager		SRC040 · SRC014
REQ060	Outside manufacture	Product may be made or packed by an outside manufacturer, who buys the materials and delivers finished product to the organisation's specification.	Procurement lead		SRC023 · SRC024
REQ061	Outside manufacture	What the organisation buys becomes its own when it arrives; what a supplier or outside manufacturer holds before then is theirs.	Finance lead		SRC023 · SRC024
REQ062	Outside manufacture	Outside manufacturers are held to the same good practice as the organisation's own plants, and their records must be available when needed.	Quality manager		SRC022
REQ063	Outside manufacture	How far back a product can be traced depends on the supplier's own records once it is outside the organisation's sight.	Quality manager		SRC009 · SRC011
REQ064	People	People are qualified to do particular work, and qualifications expire.	Plant manager		SRC001 · SRC022
REQ065	Place	A site, a warehouse and a storage place are all places, and they sit inside one another.	Supply chain lead		SRC001 · SRC039
REQ066	Place	Where something is stored affects how long it lasts and whether it stays fit to use.	Quality manager		SRC020
REQ067	Place	A place may be identified differently by the organisation, a supplier and a regulator.	Data steward		SRC011
REQ025	Process	Making a product is a sequence of steps, and the order matters as much as the ingredients.	Technical lead		SRC002 · SRC003
REQ026	Process	A step needs particular equipment, particular people and particular materials.	Plant manager		SRC002 · SRC001
REQ027	Process	The same step is used in making many different products.	Technical lead		SRC003 · SRC004
REQ028	Process	A step has settings — a temperature, a time, a speed — and the values differ by product and by line.	Technical lead		SRC003 · SRC004
REQ029	Process	A formulation is written once and made at several sites on several lines, and what is carried out differs at each.	Technical lead		SRC002 · SRC003

ID	Category	Business requirement	Owner	Subgroup	Sources
REQ030	Process	The instructions issued for one batch are a specific thing, separate from the formulation they came from.	Plant manager		SRC003 · SRC033
REQ031	Process	The order in which products run on a line matters, because what ran before can contaminate what runs next.	Quality manager	Food and beverage	SRC034
REQ032	Process	Changing a line from one product to another takes time and may need cleaning, and both are part of the cost of making.	Plant manager		SRC034 · SRC017
REQ019	Product identity	The same thing is sold in several sizes and pack formats, related to each other but made and counted separately.	Commercial lead		SRC030 · SRC031
REQ020	Product identity	Some changes make a product a different thing — a change to its declared ingredients, net content, certification or pack count — and others do not.	Regulatory lead		SRC030 · SRC031
REQ021	Product identity	What a product is made from changes over time while it remains the same product.	Technical lead		SRC030 · SRC002
REQ022	Product identity	What was actually made is a fact about the batch; what should have been made is a fact about the definition.	Quality manager		SRC005 · SRC033
REQ023	Product identity	Some things are identified individually and some only by the batch they came from, and one product may contain both.	Quality manager		SRC007 · SRC037
REQ024	Product identity	A pack may carry its own serial number as well as its batch, and retailers increasingly expect to read it, though no law requires it.	Commercial lead		SRC037 · SRC038
REQ040	Production	Making something is instructed, carried out and recorded, and the three may differ.	Plant manager		SRC033 · SRC005 · SRC021 · SRC027
REQ041	Production	Who carried out each step of a batch, and whether they were qualified to, must be recoverable.	Plant manager		SRC006 · SRC001 · SRC033
REQ042	Production	Goods move between plants, warehouses and storage places, and each movement has a time and a place.	Supply chain lead		SRC009 · SRC040
REQ044	Quality	A material or product must meet a stated specification, and what it must be changes over time.	Quality manager		SRC020 · SRC012
REQ045	Quality	Meeting a specification is shown by testing, and a test has a method as well as a result.	Quality manager		SRC020 · SRC022

ID	Category	Business requirement	Owner	Subgroup	Sources
REQ046	Quality	A departure from a procedure and a failure against a specification are different things, and both must be recorded.	Quality manager		SRC012 · SRC013
REQ047	Quality	Material that fails must be held back, a decision made about it, and that decision made by someone with the authority to make it.	Quality manager		SRC014 · SRC013
REQ048	Quality	Every departure from procedure must be investigated, with conclusions and follow-up recorded.	Quality manager		SRC015 · SRC013
REQ049	Quality	A significant or recurring problem needs its cause found, action taken, and the action shown to have worked before it is closed.	Quality manager		SRC014 · SRC013
REQ050	Quality	Output is not available to sell until someone releases it, and release is a decision someone makes.	Quality manager		SRC005 · SRC003
REQ051	Quality	When a batch is implicated in a food safety issue, the records of what went into it and where it went must be produced within twenty-four hours.	Quality manager	Food and beverage	SRC009 · SRC010
REQ052	Quality	Traceability records must be kept for two years.	Quality manager	Food and beverage	SRC011
REQ053	Quality	A lot code is assigned when a product is first packed or transformed, and must be carried at every later movement.	Quality manager	Food and beverage	SRC011 · SRC009
REQ054	Quality	A complaint may concern a specific batch, and that link is what connects experience of the product to how it was made.	Quality manager		SRC022 · SRC005
REQ055	Quality	Some records must be signed, and a signature must show who signed, what, when and what the signature meant.	Compliance officer		SRC032 · SRC033
REQ056	Quality	Once recorded, an entry stays visible and unaltered; a correction is a new entry, not an edit.	Compliance officer		SRC032
REQ057	Quality	Some steps must be checked by a second qualified person or by approved equipment.	Quality manager		SRC033
REQ058	Supplier	A supplier is assessed before being used and reassessed afterwards.	Procurement lead		SRC020 · SRC029
REQ059	Supplier	A problem with a supplier is raised with them formally and their response followed up.	Procurement lead		SRC014 · SRC020

4 Data Requirements

ID	Must record	Must answer	As at	Must prevent	Business Requirements	Prevented by
DRQ001	Every recipe line: the product made, the input, and how much of the input goes into one of the product The version of the recipe each line belongs to	What goes into this product, and how much of each input ends up in one finished unit?	both	A product that contains itself, however many steps away A recipe read from lines belonging to two different versions	REQ002 · REQ003 · REQ005	composition — REL001 parent-cardinality — SBJ015
DRQ002	The same recipe lines, read from the input upwards	Which finished products contain this ingredient, by any route?	current	—	REQ003 · REQ005	—
DRQ003	Use of an input as a whole, never of what lies inside it	Which products use this intermediate, and are any using a variant of it that should be its own product?	current	Adjusting what goes into an input from inside the product that uses it	REQ004	composition — REL001
DRQ004	The quantity the recipe specified for each input The quantity each batch actually used	How much more or less of each input did this batch use than the recipe specified?	as-at	—	REQ006 · REQ022	—
DRQ005	Which input actually filled each place in the recipe on each batch Approved alternatives and the conditions for using them	Which batches used an alternative input, and for which ingredient?	as-at	An alternative used without approval	REQ007	mandatory-attribute — ATR042
DRQ006	Each supplier's approval for each material, with the dates it held Each assessment of a supplier, its scope and how long it is valid	Was this supplier approved for this material on the day it was received?	as-at	An approval treated as current after it lapsed	REQ008 · REQ058	dates — REL003
DRQ007	What a supplier's document states about a batch, who issued it, and whether the organisation checked it	Which of this batch's stated properties were tested here, and which were taken on the supplier's word?	current	A supplier's claim reported as though it were the organisation's own test	REQ009	mandatory-attribute — ATR014

ID	Must record	Must answer	As at	Must prevent	Business Requirements	Prevented by
DRQ008	Which inputs are bought in finished, with contents the organisation does not know	Which products depend on a bought-in ingredient whose contents we rely on the supplier to declare?	current	—	REQ010 · REQ063	—
DRQ009	The allergens, other hazards, certifications and origin of each input, and which rules apply to each kind of product	What allergens, hazards and certifications does this product carry, and from which inputs?	both	A product declared free of something one of its inputs contains	REQ011 · REQ012 · REQ014	composition — REL001
DRQ010	Each ingredient of a household product, how much of it there is, and whether it is withheld as confidential	What must be disclosed for this household product, and what is withheld?	current	—	REQ013	—
DRQ011	The use-by date of each batch, and the inputs that set it	What is the latest date this finished batch can be used, given what went into it?	current	—	REQ015	—
DRQ012	Every split and every combination of batches, with the quantities of each part	Which batches came from this one, and which batches made up this combined one?	current	A split that produces a quantity with no batch it came from	REQ016	relationship-cardinality — REL001
DRQ013	How much of each item is held, in what condition, at which place, at what time	How much of this material is available, on hold, or rejected at each site right now?	both	Stock on hold counted as available	REQ001 · REQ017 · REQ018	parent-cardinality — SBJ048
DRQ014	Each size and pack as its own product, related to its family Each change to a product, and whether it made a new product or a new version, and why	Which pack sizes of this product exist, and did this change create a new product or a new version?	as-at	—	REQ019 · REQ020	—
DRQ015	The versions of each recipe, when each took effect, and which version each batch was made to	What recipe was this batch made to, and what was it meant to contain at the time?	as-at	—	REQ021 · REQ022	—
DRQ016	Each individually identified unit, the product it is a unit of, and which unit or batch filled each part of it	What exactly is inside this individual unit, and which batch did each part come from?	as-at	A unit recorded with a part its definition does not call for	REQ023 · REQ024	relationship-cardinality — REL015

ID	Must record	Must answer	As at	Must prevent	Business Requirements	Prevented by
DRQ017	The steps of each method in order, what each step needs, and its settings for each product on each line	What steps, equipment, qualifications and settings does making this product need on this line?	current	—	REQ025 · REQ026 · REQ027 · REQ028	—
DRQ018	The four levels of each recipe, and which level each came from	Which instructions were issued for this batch, and which formulation did they come from?	as-at	—	REQ029 · REQ030	—
DRQ019	The rules about what may run after what on each line, and why Each changeover and each clean, with how long it took	What ran on this line just before this batch, and was the line cleaned in between?	as-at	A product free of an allergen run straight after one containing it, with no clean between	REQ031 · REQ032	mandatory-attribute — ATR058
DRQ020	Each place in the plant and what it sits within Which machine stood in each place, from when to when	Which machine was in this place on the day this batch was made?	as-at	—	REQ033 · REQ034	—
DRQ021	Each machine's failures and how it failed, its repairs, its readings, and how much it matters	Which machines cause the most unplanned stoppage, and how do they fail?	both	—	REQ035 · REQ036 · REQ039	—
DRQ022	One record of each stoppage — start, end, and reason from a set list — with the lost output and the repair both tied to it	How long did this line stop, why, and what did maintenance do about it?	as-at	One stoppage recorded twice A stoppage with no reason	REQ037 · REQ038	relationship-cardinality — REL031 lookup — LKP006
DRQ023	What each step of a batch was told to do, what it did, what resulted, who did it, and whether they were qualified at the time	Who carried out each step of this batch, and were they qualified on that day?	as-at	Work by someone not qualified going unnoticed	REQ040 · REQ041 · REQ064	dates — REL021
DRQ024	Each movement of goods — from where, to where, when, how much, which batch	Where has this batch been, and where is it now?	both	—	REQ042 · REQ065	—

ID	Must record	Must answer	As at	Must prevent	Business Requirements	Prevented by
DRQ025	Each version of a specification, and each test with its method, sample, sampling plan and result	Did this batch meet the specification in force when it was made, and by what test?	as-at	—	REQ044 · REQ045	—
DRQ026	Departures from procedure against the batch run, and failures against specification against the batch made — separately, each with how serious it is and how it was contained	What went wrong on this batch — a procedure not followed, the product out of specification, or both?	current	The two being recorded as one	REQ046 · REQ048	platform capability
DRQ027	Each hold, the decision taken, the reason, and who approved it	What was decided about this failed material, why, and by whom?	current	A decision with no named approver	REQ047	mandatory-attribute — ATR010
DRQ028	For each corrective action: the cause, what was done, the check that it worked, and when it was closed	Which corrective actions are open, and were any closed without checking they worked?	current	Closing an action before checking it worked	REQ049 · REQ059	mandatory-attribute — ATR113
DRQ029	Each release — who released it, when, on what basis, and what it covered	Who released this batch, when, and on what basis?	current	Stock treated as available because a batch finished	REQ050	mandatory-attribute — ATR115
DRQ030	The lot code given at first packing or transformation, carried at every later movement, kept for two years	Every batch this ingredient batch went into, and everywhere those went — within twenty-four hours	current	A movement recorded without its lot code	REQ051 · REQ052 · REQ053	mandatory-attribute — ATR029
DRQ031	Each complaint, and the batch where the person can supply it	Which complaints concern this batch, and what else was made from the same inputs?	current	—	REQ054	—
DRQ032	Each signature — who, what, when, and what it meant — each second check, and each correction as a new entry	Who signed this record, what did it mean, and has anything been corrected since?	as-at	An entry altered without a correction being recorded	REQ055 · REQ056 · REQ057	relationship-cardinality — REL035

ID	Must record	Must answer	As at	Must prevent	Business Requirements	Prevented by
DRQ033	Which party made or packed each batch, the specification they were given, and when the goods became the organisation's	Which of our finished batches were made by an outside manufacturer, to which specification, and when did they become ours?	as-at	—	REQ060 · REQ061 · REQ062	—
DRQ034	The conditions each place must keep, and any breach against the stock held there	Was this batch ever stored outside its permitted conditions?	as-at	—	REQ066	—
DRQ035	The identifiers each place and each product carries, and who issued each	Which place or product is this, whatever identifier the supplier or regulator used?	current	—	REQ067 · REQ069	—
DRQ036	The expected and actual cost of each product, by element	What did this batch cost against its standard, and why?	as-at	—	REQ070	—

5 Data Model

5.1 Canonical Roles

One row per role. A role is canonical in its own right; the pairs it forms live in the relationships that use it.

Domain	Role ID	Role	Definition	Established by	Also called	Requirements
Documents	ROL067	Declaring Document (Declaring Documents)	A document is a Declaring Document while it declares properties of a product batch, which is its Declared Batch.	REL037	—	REQ009 · REQ010
Equipment	ROL017	Approved Position (Approved Positions)	A place in the plant is an Approved Position while it is approved for a step, which is its Approved Step.	REL008	—	REQ026 · REQ029
Equipment	ROL020	Contained Position (Contained Positions)	A place in the plant is a Contained Position while it is within another, which is its Containing Position.	REL010	—	REQ033
Equipment	ROL021	Containing Position (Containing Positions)	A place in the plant is a Containing Position while it contains another, which is its Contained Position.	REL010	—	REQ033
Equipment	ROL022	Occupied Position (Occupied Positions)	A place in the plant is an Occupied Position while it is occupied by a machine, which is its Installed Machine.	REL011	—	REQ034
Equipment	ROL047	Running Position (Running Positions)	A place in the plant is a Running Position while a run ran at it, which is its Production Run.	REL026	—	REQ040 · REQ034
Equipment	ROL024	Sited Position (Sited Positions)	A place in the plant is a Sited Position while it is sited at a place, which is its Housing Site.	REL012	—	REQ033 · REQ065
Events	ROL059	Complaint (Complaints)	An occurrence is a Complaint while it reports a problem with a product batch, which is its Complained-of Batch.	REL033	—	REQ054
Events	ROL051	Consuming Run (Consuming Runs)	An occurrence is a Consuming Run while it consumed a product batch, which is its Input Batch.	REL029	—	REQ006 · REQ022
Events	ROL064	Corrected Record (Corrected Records)	An occurrence is a Corrected Record while a correction corrects it, which is its Correction.	REL035	—	REQ056

Domain	Role ID	Role	Definition	Established by	Also called	Requirements
Events	ROL063	Correction (Corrections)	An occurrence is a Correction while it corrects a record, which is its Corrected Record.	REL035	—	REQ056
Events	ROL057	Finding (Findings)	An occurrence is a Finding while it is raised against a product batch, which is its Affected Batch.	REL032	—	REQ046 · REQ047
Events	ROL049	Performed Work (Performed Works)	An occurrence is Performed Work while it was performed by a person, who is its Performer.	REL028	—	REQ041 · REQ047 · REQ050 · REQ057
Events	ROL053	Producing Run (Producing Runs)	An occurrence is a Producing Run while it produced a product batch, which is its Output Batch.	REL030	—	REQ022
Events	ROL046	Production Run (Production Runs)	An occurrence is a Production Run while it ran at a place in the plant and under a recipe, which are its Running Position and Issued Instructions.	REL026 · REL027	—	REQ040
Events	ROL061	Response (Responses)	An occurrence is a Response while it responds to another occurrence, which is its Trigger.	REL034	—	REQ049
Events	ROL055	Stoppage (Stoppages)	An occurrence is a Stoppage while it stopped a machine, which is its Stopped Machine.	REL031	—	REQ037 · REQ038
Events	ROL069	Storage Breach (Storage Breaches)	An occurrence is a Storage Breach while it exposed a product batch, which is its Exposed Batch.	REL038	—	REQ066
Events	ROL065	Supplier Finding (Supplier Findings)	An occurrence is a Supplier Finding while it is raised with an organisation, which is its Responsible Supplier.	REL036	—	REQ059
Events	ROL062	Trigger (Triggers)	An occurrence is a Trigger while it prompted another occurrence, which is its Response.	REL034	—	REQ049
Instances	ROL028	Assembled Unit (Assembled Units)	A unit is an Assembled Unit while it is built from another unit or incorporates a batch, which are its Fitted Unit and Incorporated Batch.	REL014 · REL015	—	REQ023
Instances	ROL029	Fitted Unit (Fitted Units)	A unit is a Fitted Unit while it is built into another unit, which is its Assembled Unit.	REL014	—	REQ023

Domain	Role ID	Role	Definition	Established by	Also called	Requirements
Instances	ROL026	Individual Unit (Individual Units)	A unit is an Individual Unit while it is a unit of a product, which is its Defining Product.	REL013	—	REQ023 · REQ024
Instances	ROL023	Installed Machine (Installed Machines)	A machine is an Installed Machine while it occupies a place in the plant, which is its Occupied Position.	REL011	—	REQ034 · REQ035
Instances	ROL056	Stopped Machine (Stopped Machines)	A machine is a Stopped Machine while a stoppage stopped it, which is its Stoppage.	REL031	—	REQ035 · REQ037
Legal Entities	ROL006	Approved Supplier (Approved Suppliers)	An organisation is an Approved Supplier while it may supply a product, which is its Sourced Material.	REL003	—	REQ008 · REQ058
Legal Entities	ROL009	Certifying Body (Certifying Bodies)	An organisation is a Certifying Body while it certifies a product or a place, which is its Certified Product or Certified Site.	REL004 · REL017	—	REQ011
Legal Entities	ROL038	Employer (Employers)	An organisation is an Employer while a person works for it, who is its Employee.	REL020	—	REQ041
Legal Entities	ROL011	Outside Manufacturer (Outside Manufacturers)	An organisation is an Outside Manufacturer while it manufactures a product for this organisation, which is its Contracted Product.	REL005	—	REQ060 · REQ062
Legal Entities	ROL036	Registering Authority (Registering Authorities)	An organisation is a Registering Authority while it registers a place, which is its Registered Place.	REL018	—	REQ067
Legal Entities	ROL066	Responsible Supplier (Responsible Suppliers)	An organisation is a Responsible Supplier while a finding is raised with it, which is its Supplier Finding.	REL036	—	REQ059
Legal Entities	ROL033	Site Operator (Site Operators)	An organisation is a Site Operator while it operates a place, which is its Operated Site.	REL019	—	REQ060 · REQ062
Legal Entities	ROL042	Trading Party (Trading Parties)	An organisation is a Trading Party while an exchange is transacted with it, which is its Recorded Exchange.	REL022	—	REQ061

Domain	Role ID	Role	Definition	Established by	Also called	Requirements
Locations	ROL008	Certified Site (Certified Sites)	A place is a Certified Site while it is certified by an organisation, which is its Certifying Body.	REL017	—	REQ011
Locations	ROL031	Contained Place (Contained Places)	A place is a Contained Place while it is within another place, which is its Containing Place.	REL016	—	REQ065
Locations	ROL032	Containing Place (Containing Places)	A place is a Containing Place while it contains another place, which is its Contained Place.	REL016	—	REQ065
Locations	ROL044	Despatching Place (Despatching Places)	A place is a Despatching Place while an exchange ships from it, which is its Recorded Exchange.	REL024	—	REQ042
Locations	ROL025	Housing Site (Housing Sites)	A place is a Housing Site while it houses a place in the plant, which is its Sited Position.	REL012	—	REQ065
Locations	ROL034	Operated Site (Operated Sites)	A place is an Operated Site while it is operated by an organisation, which is its Site Operator.	REL019	—	REQ062
Locations	ROL045	Receiving Place (Receiving Places)	A place is a Receiving Place while an exchange is delivered to it, which is its Recorded Exchange.	REL025	—	REQ042
Locations	ROL035	Registered Place (Registered Places)	A place is a Registered Place while it is registered with an organisation, which is its Registering Authority.	REL018	—	REQ067
People	ROL037	Employee (Employees)	A person is an Employee while they work for an organisation, which is their Employer.	REL020	—	REQ041
People	ROL050	Performer (Performers)	A person is a Performer while they performed an occurrence, which is their Performed Work.	REL028	—	REQ041 · REQ047 · REQ050
People	ROL039	Qualified Person (Qualified People)	A person is a Qualified Person while they are qualified for a step of work, which is their Qualified Work.	REL021	—	REQ064 · REQ041
Processes	ROL016	Approved Step (Approved Steps)	A step is an Approved Step while it is approved at a place in the plant, which is its Approved Position.	REL008	—	REQ026 · REQ029
Processes	ROL018	Derived Recipe (Derived Recipes)	A recipe is a Derived Recipe while it is derived from another recipe, which is its Source Recipe.	REL009	—	REQ029 · REQ030

Domain	Role ID	Role	Definition	Established by	Also called	Requirements
Processes	ROL015	Earlier Step (Earlier Steps)	A step is an Earlier Step while it precedes another step, which is its Later Step.	REL007	—	REQ025
Processes	ROL048	Issued Instructions (Issued Instructions)	A recipe is Issued Instructions while a run ran under it, which is its Production Run.	REL027	—	REQ030
Processes	ROL014	Later Step (Later Steps)	A step is a Later Step while it follows another step, which is its Earlier Step.	REL007	—	REQ025
Processes	ROL013	Making Method (Making Methods)	A method is a Making Method while it makes a product, which is its Routed Product.	REL006	—	REQ025 REQ027
Processes	ROL040	Qualified Work (Qualified Works)	A step is Qualified Work while it may be performed by a person, who is its Qualified Person.	REL021	—	REQ064
Processes	ROL019	Source Recipe (Source Recipes)	A recipe is a Source Recipe while it is the source of another recipe, which is its Derived Recipe.	REL009	—	REQ029
Products	ROL058	Affected Batch (Affected Batches)	A product batch is an Affected Batch while a finding is raised against it, which is its Finding.	REL032	—	REQ046
Products	ROL007	Certified Product (Certified Products)	A product is a Certified Product while it is certified by an organisation, which is its Certifying Body.	REL004	—	REQ011
Products	ROL060	Complained-of Batch (Complained-of Batches)	A product batch is a Complained-of Batch while a complaint reports a problem with it, which is its Complaint.	REL033	—	REQ054
Products	ROL010	Contracted Product (Contracted Products)	A product is a Contracted Product while it is manufactured by an organisation, which is its Outside Manufacturer.	REL005	—	REQ060
Products	ROL068	Declared Batch (Declared Batches)	A product batch is a Declared Batch while a document declares its properties, which is its Declaring Document.	REL037	—	REQ009
Products	ROL027	Defining Product (Defining Products)	A product is a Defining Product while individually identified units are made to it, which are its Individual Units.	REL013	—	REQ023

Domain	Role ID	Role	Definition	Established by	Also called	Requirements
Products	ROL070	Exposed Batch (Exposed Batches)	A product batch is an Exposed Batch while a storage breach exposed it, which is its Storage Breach.	REL038	—	REQ066
Products	ROL004	Family (Families)	A product is a Family while it has variant products, which are its Variants.	REL002	—	REQ019
Products	ROL030	Incorporated Batch (Incorporated Batches)	A product batch is an Incorporated Batch while it is incorporated into a unit, which is its Assembled Unit.	REL015	—	REQ023
Products	ROL052	Input Batch (Input Batches)	A product batch is an Input Batch while a run consumed it, which is its Consuming Run.	REL029	—	REQ022 · REQ051
Products	ROL001	Input Material (Input Materials)	A product is an Input Material while it is used in another product, which is its Output Product.	REL001	—	REQ002 · REQ003 · REQ005
Products	ROL054	Output Batch (Output Batches)	A product batch is an Output Batch while a run produced it, which is its Producing Run.	REL030	—	REQ022 · REQ053
Products	ROL002	Output Product (Output Products)	A product is an Output Product while it is made from another product, which is its Input Material.	REL001	—	REQ002 · REQ004
Products	ROL012	Routed Product (Routed Products)	A product is a Routed Product while it is made by a method, which is its Making Method.	REL006	—	REQ025
Products	ROL005	Sourced Material (Sourced Materials)	A product is a Sourced Material while it may be sourced from an organisation, which is its Approved Supplier.	REL003	—	REQ008
Products	ROL043	Traded Product (Traded Products)	A product is a Traded Product while an exchange concerns it, which is its Recorded Exchange.	REL023	—	REQ042
Products	ROL003	Variant (Variants)	A product is a Variant while it is a variant within a family product, which is its Family.	REL002	—	REQ019
Transactions	ROL041	Recorded Exchange (Recorded Exchanges)	An exchange is a Recorded Exchange while it is transacted with an organisation, in a place, or concerns a product — its Trading Party, Despatching or Receiving Place, and Traded Product.	REL022 · REL023 · REL024 · REL025	—	REQ042 · REQ061

5.2 Role–Verb–Role Relationships

Each row reads as a sentence: source domain as source role, forward verb, destination domain as target role.
Relationship attributes are in Appendix F.

ID	Source domain	Source role	Forward verb	Target role	Destination	Inverse verb	Forward view	Mirror view	Cardinality	Dates	DDA
REL037	Documents	ROL067 Declaring Document	declares properties of	ROL068 Declared Batch	Products	is declared by	Document Declared Batches	Product DeclaringDocuments	MANY_TO_MANY	no	create
REL010	Equipment	ROL020 Contained Position	is within	ROL021 Containing Position	Equipment	contains	EquipmentPositionContaining Positions	Equipment Position ContainedPositions	MANY_TO_ONE	no	create
REL011	Equipment	ROL022 Occupied Position	is occupied by	ROL023 Installed Machine	Instances	occupies	EquipmentPositionInstalled Machines	InstanceOccupiedPositions	MANY_TO_MANY	yes	create
REL012	Equipment	ROL024 Sited Position	is sited at	ROL025 Housing Site	Locations	houses	EquipmentPositionHousingSites	LocationSited Positions	MANY_TO_ONE	no	create
REL033	Events	ROL059 Complaint	reports a problem with	ROL060 Complained-of Batch	Products	is complained about in	Event Complained-of Batches	Product Complaints	MANY_TO_ONE	no	create
REL029	Events	ROL051 Consuming Run	consumed	ROL052 Input Batch	Products	was consumed by	EventInput Batches	Product Consuming Runs	MANY_TO_MANY	no	create
REL035	Events	ROL063 Correction	corrects	ROL064 Corrected Record	Events	is corrected by	EventCorrected Records	EventCorrections	MANY_TO_ONE	no	create
REL032	Events	ROL057 Finding	is raised against	ROL058 Affected Batch	Products	is subject of	EventAffected Batches	ProductFindings	MANY_TO_MANY	no	create
REL028	Events	ROL049 Performed Work	was performed by	ROL050 Performer	People	performed	EventPerformers	Person Performed Works	MANY_TO_MANY	yes	create
REL030	Events	ROL053 Producing Run	produced	ROL054 Output Batch	Products	was produced by	EventOutput Batches	ProductProducingRuns	MANY_TO_MANY	no	create

ID	Source domain	Source role	Forward verb	Target role	Destination	Inverse verb	Forward view	Mirror view	Cardinality	Dates	DDA
REL026	Events	ROL046 Production Run	ran at	ROL047 Running Position	Equipment	ran	EventRunning Positions	Equipment Position Production Runs	MANY_TO_ONE	no	create
REL027	Events	ROL046 Production Run	ran under	ROL048 Issued Instructions	Processes	governed	EventIssued Instructions	Process Segment Production Runs	MANY_TO_ONE	no	create
REL034	Events	ROL061 Response	responds to	ROL062 Trigger	Events	prompted	EventTriggers	Event Responses	MANY_TO_MANY	no	create
REL031	Events	ROL055 Stoppage	stopped	ROL056 Stopped Machine	Instances	was stopped in	EventStopped Machines	InstanceStop-pages	MANY_TO_ONE	no	create
REL038	Events	ROL069 Storage Breach	exposed	ROL070 Exposed Batch	Products	was exposed in	EventExposed Batches	ProductStorage Breaches	MANY_TO_MANY	no	create
REL036	Events	ROL065 Supplier Finding	is raised with	ROL066 Responsible Supplier	Legal Entities	received	EventResponsibleSuppliers	LegalEntity SupplierFindings	MANY_TO_ONE	yes	create
REL014	Instances	ROL028 Assembled Unit	is built from	ROL029 Fitted Unit	Instances	is built into	InstanceFitted Units	Instance Assembled Units	MANY_TO_MANY	yes	create
REL015	Instances	ROL028 Assembled Unit	incorporates batch	ROL030 Incorporated Batch	Products	is incorporated into	InstanceIncorporatedBatches	Product Assembled Units	MANY_TO_MANY	no	create
REL013	Instances	ROL026 Individual Unit	is a unit of	ROL027 Defining Product	Products	is made as	InstanceDefining Products	ProductIndividualUnits	MANY_TO_ONE	no	create
REL019	Legal Entities	ROL033 Site Operator	operates	ROL034 Operated Site	Locations	is operated by	LegalEntityOperatedSites	LocationSite Operators	MANY_TO_MANY	yes	create

ID	Source domain	Source role	Forward verb	Target role	Destination	Inverse verb	Forward view	Mirror view	Cardinality	Dates	DDA
REL017	Locations	ROL008 Certified Site	is certified by	ROL009 Certifying Body	Legal Entities	certifies	LocationCertifyingBodies	LegalEntity CertifiedSites	MANY_TO_MANY	yes	create
REL016	Locations	ROL031 Contained Place	is within	ROL032 Containing Place	Locations	contains	Location Containing Places	Location Contained Places	MANY_TO_ONE	no	create
REL018	Locations	ROL035 Registered Place	is registered with	ROL036 Registering Authority	Legal Entities	registers	LocationRegisteringAuthorities	LegalEntity Registered Places	MANY_TO_MANY	yes	create
REL020	People	ROL037 Employee	works for	ROL038 Employer	Legal Entities	employs	Person Employers	LegalEntity Employees	MANY_TO_MANY	yes	create
REL021	People	ROL039 Qualified Person	is qualified for	ROL040 Qualified Work	Processes	may be performed by	PersonQualified Works	Process SegmentQualifiedPeople	MANY_TO_MANY	yes	create
REL008	Processes	ROL016 Approved Step	is approved at	ROL017 Approved Position	Equipment	is approved for	Process Segment ApprovedPositions	Equipment Position ApprovedSteps	MANY_TO_MANY	yes	create
REL009	Processes	ROL018 Derived Recipe	is derived from	ROL019 Source Recipe	Processes	is the source of	Process SegmentSource Recipes	Process Segment Derived Recipes	MANY_TO_ONE	yes	create
REL007	Processes	ROL014 Later Step	follows	ROL015 Earlier Step	Processes	precedes	Process SegmentEarlier Steps	Process SegmentLater Steps	MANY_TO_MANY	no	create
REL004	Products	ROL007 Certified Product	is certified by	ROL009 Certifying Body	Legal Entities	certifies	ProductCertifyingBodies	LegalEntity Certified Products	MANY_TO_MANY	yes	create
REL005	Products	ROL010 Contracted Product	is manufactured by	ROL011 Outside Manufacturer	Legal Entities	manufactures	ProductOutside Manufacturers	LegalEntity Contracted Products	MANY_TO_MANY	yes	create

ID	Source domain	Source role	Forward verb	Target role	Destination	Inverse verb	Forward view	Mirror view	Cardinality	Dates	DDA
REL001	Products	ROL002 Output Product	is made from	ROL001 Input Material	Products	is used in	ProductInput Materials	ProductOutput Products	MANY_TO_MANY	yes	create
REL006	Products	ROL012 Routed Product	is made by	ROL013 Making Method	Processes	makes	ProductMaking Methods	Process Segment Routed Products	MANY_TO_ONE	yes	create
REL003	Products	ROL005 Sourced Material	may be sourced from	ROL006 Approved Supplier	Legal Entities	may supply	Product Approved Suppliers	LegalEntity SourcedMaterials	MANY_TO_MANY	yes	create
REL002	Products	ROL003 Variant	is a variant within	ROL004 Family	Products	has variant	ProductFamilies	ProductVariants	MANY_TO_ONE	yes	create
REL022	Transactions	ROL041 Recorded Exchange	is transacted with	ROL042 Trading Party	Legal Entities	transacts	Transaction TradingParties	LegalEntity Recorded Exchanges	MANY_TO_MANY	no	create
REL023	Transactions	ROL041 Recorded Exchange	concerns	ROL043 Traded Product	Products	is traded in	Transaction TradedProducts	Product Recorded Exchanges	MANY_TO_MANY	no	create
REL024	Transactions	ROL041 Recorded Exchange	ships from	ROL044 Despatching Place	Locations	despatches	Transaction Despatching Places	Location Recorded Exchanges	MANY_TO_ONE	no	create
REL025	Transactions	ROL041 Recorded Exchange	is delivered to	ROL045 Receiving Place	Locations	receives	Transaction ReceivingPlaces	Location Recorded Exchanges	MANY_TO_ONE	no	create

5.3 Concepts

Every concept in scope, whether DDA already holds it or this model proposes it. Each definition states the grain — one record per what. Concept attributes are in Appendix G.

Domain	Concept ID	Concept	Action	Definition	Parent	Cardinality to parent	Kind	Mastery	From
Documents	SBJ010	Document	reuse DMC000000142	One document — a certificate of analysis, a specification sheet, a safety data sheet. One record per document.	—	—	subject	ingested	DRQ007
Documents	SBJ011	Document Declarations	create	One property a document states about a batch or product, and whether the organisation verified it. One record per document per property.	Document	ONE_TO_MANY_OPTIONAL	subject	ingested	DRQ007 · DRQ008
Equipment	SBJ031	EquipmentPosition	create	One place in a plant where work is performed — a site level, an area, a work centre, a work unit, an equipment module. One record per place in the plant structure.	—	—	subject	mastered	DRQ020
Equipment	SBJ033	Equipment Capabilities	create	One thing a place is able to do, and how much of it. One record per place per capability.	Equipment Position	ONE_TO_MANY_OPTIONAL	subject	mastered	DRQ017
Equipment	SBJ035	EquipmentMaintenancePlans	create	One planned maintenance regime for a place, and what triggers it. One record per place per plan.	Equipment Position	ONE_TO_MANY_OPTIONAL	subject	mastered	DRQ021
Equipment	SBJ032	EquipmentPositionIdentifiers	create	One identifier by which a place is known, in one scheme. One record per place per scheme per identifier.	Equipment Position	ONE_TO_MANY_OPTIONAL	subject	mastered	DRQ020
Equipment	SBJ034	Equipment Sequencing Rules	create	One constraint on what may run after what at a place, with the reason. One record per place per constraint.	Equipment Position	ONE_TO_MANY_OPTIONAL	subject	mastered	DRQ019
Events	SBJ049	Event	create	One thing that happened in the operation, at a time and a place. One record per occurrence.	—	—	occurrence	ingested	DRQ023

Domain	Concept ID	Concept	Action	Definition	Parent	Cardinality to parent	Kind	Mastery	From
Events	SBJ053	EventBatch Splits	create	One batch divided into parts, or several combined into one. One record per split or combination.	Event	ONE_TO_MANY_OPTIONAL	occurrence	ingested	DRQ012
Events	SBJ054	Event Changeovers	create	One change of a place from one product to another, including any clean. One record per changeover.	Event	ONE_TO_MANY_OPTIONAL	occurrence	ingested	DRQ019
Events	SBJ064	Event Complaints	create	One report from outside the organisation about a product. One record per complaint.	Event	ONE_TO_MANY_OPTIONAL	occurrence	ingested	DRQ031
Events	SBJ065	EventCorrections	create	One correction to something already recorded, replacing it without erasing it. One record per correction.	Event	ONE_TO_MANY_OPTIONAL	occurrence	mastered	DRQ032
Events	SBJ061	EventCorrective Actions	create	One response to a problem — its cause, what was done, the check that it worked, and its closure. One record per corrective action.	Event	ONE_TO_MANY_OPTIONAL	occurrence	mastered	DRQ028
Events	SBJ058	EventDeviations	create	One departure from a defined procedure during a run. One record per departure.	Event	ONE_TO_MANY_OPTIONAL	occurrence	mastered	DRQ026
Events	SBJ060	EventDispositions	create	One decision about failed material, with its reason and its approver. One record per decision.	Event	ONE_TO_MANY_OPTIONAL	occurrence	mastered	DRQ027
Events	SBJ056	EventMaintenanceWork	create	One piece of maintenance work — what prompted it, what was found, what was done. One record per piece of work.	Event	ONE_TO_MANY_OPTIONAL	occurrence	ingested	DRQ021 · DRQ022
Events	SBJ052	EventMaterial Consumptions	create	One quantity of one input batch used in one run, filling one place in the recipe. One record per run per input batch per recipe position.	Event	ONE_TO_MANY_OPTIONAL	occurrence	ingested	DRQ004 · DRQ005

Domain	Concept ID	Concept	Action	Definition	Parent	Cardinality to parent	Kind	Mastery	From
Events	SBJ059	EventNon Conformances	create	One failure of a batch or material against its specification. One record per failure.	Event	ONE_TO_MANY_OPTIONAL	occurrence	mastered	DRQ026
Events	SBJ050	EventProduction Orders	create	One instruction to make a quantity of a product. One record per order.	Event	ONE_TO_MANY_OPTIONAL	occurrence	ingested	DRQ023
Events	SBJ051	EventProduction Runs	create	One batch actually made — what was instructed, what was done, and what came out. One record per batch run.	Event	ONE_TO_MANY_OPTIONAL	occurrence	ingested	DRQ004 · DRQ023
Events	SBJ063	EventRecalls	create	One safety or quality incident, its scope and its resolution. One record per incident.	Event	ONE_TO_MANY_OPTIONAL	occurrence	mastered	DRQ030
Events	SBJ062	EventReleases	create	One act of making output available, by one person, on a stated basis. One record per release.	Event	ONE_TO_MANY_OPTIONAL	occurrence	mastered	DRQ029
Events	SBJ055	EventStoppages	create	One stop in production at one place — start, end, and why. One record per stoppage.	Event	ONE_TO_MANY_OPTIONAL	occurrence	ingested	DRQ022
Events	SBJ066	EventStorage Breaches	create	One period a place failed to keep its required conditions. One record per breach.	Event	ONE_TO_MANY_OPTIONAL	occurrence	ingested	DRQ034
Events	SBJ057	EventTests	create	One test against a specification — its method, its sample and its result. One record per test.	Event	ONE_TO_MANY_OPTIONAL	occurrence	ingested	DRQ025
Instances	SBJ026	Instance	create	One individually identified physical thing — a serialised pack or a machine — made to a product definition. One record per individually identified unit.	—	—	subject	mastered	DRQ016 · DRQ020

Domain	Concept ID	Concept	Action	Definition	Parent	Cardinality to parent	Kind	Mastery	From
Instances	SBJ028	InstanceConditions	create	One reading of a unit's state — hours run, cycles, wear, calibration result. One record per unit per reading.	Instance	ONE_TO_MANY_OPTIONAL	occurrence	ingested	DRQ021
Instances	SBJ030	InstanceCriticalities	create	One assessment of how much a machine matters — to output, to safety, and whether a spare exists. One record per machine per assessment.	Instance	ONE_TO_MANY_OPTIONAL	subject	mastered	DRQ021
Instances	SBJ029	InstanceFailure Modes	create	One way a machine characteristically fails, and what that failure causes. One record per machine per failure mode.	Instance	ONE_TO_MANY_OPTIONAL	subject	mastered	DRQ021
Instances	SBJ027	InstanceIdentifiers	create	One identifier by which a unit is individually known, in one scheme. One record per unit per scheme per identifier.	Instance	ONE_TO_MANY_OPTIONAL	subject	mastered	DRQ016
Legal Entities	SBJ005	LegalEntity	reuse DMC000000106	One organisation — a supplier, an outside manufacturer, a certifying body, or one of the organisation's own companies. One record per organisation.	—	—	subject	ingested	DRQ006 · DRQ033
Legal Entities	SBJ007	LegalEntity Address	reuse DMC000000108	One address of an organisation. One record per organisation per address type.	LegalEntity	ONE_TO_MANY_OPTIONAL	subject	ingested	DRQ006
Legal Entities	SBJ008	LegalEntity Assessments	create	One assessment of a supplier or outside manufacturer — its scope, its result, and how long it is valid. One record per organisation per assessment.	LegalEntity	ONE_TO_MANY_OPTIONAL	subject	mastered	DRQ006
Legal Entities	SBJ006	LegalEntityIdentifiers	reuse DMC000000023	One identifier by which an organisation is known, in one scheme. One record per organisation per scheme per identifier.	LegalEntity	ONE_TO_MANY_OPTIONAL	subject	ingested	DRQ035
Legal Entities	SBJ009	LegalEntity Supplier Requests	create	One problem formally raised with a supplier, with their response and its closure. One record per request.	LegalEntity	ONE_TO_MANY_OPTIONAL	occurrence	mastered	DRQ028

Domain	Concept ID	Concept	Action	Definition	Parent	Cardinality to parent	Kind	Mastery	From
Locations	SBJ043	Location	create	One identifiable place — a site, a plant, a warehouse, a storage position. One record per place.	—	—	subject	mastered	DRQ024
Locations	SBJ044	LocationIdentifiers	create	One identifier by which a place is known, and who issued it. One record per place per scheme per identifier.	Location	ONE_TO_MANY_OPTIONAL	subject	mastered	DRQ035
Locations	SBJ045	LocationStorageConditions	create	One condition a place must keep — temperature, humidity — with its limits. One record per place per condition.	Location	ONE_TO_MANY_OPTIONAL	subject	mastered	DRQ034
People	SBJ001	Person	reuse DMC000000143	One natural person — an operator, a technician, a supplier's contact. One record per person.	—	—	subject	mastered	DRQ023
People	SBJ002	PersonQualificationRecords	create	One qualification a person holds to perform particular work, with the basis on which it was awarded. One record per person per qualification per award.	Person	ONE_TO_MANY_OPTIONAL	subject	ingested	DRQ023
People	SBJ003	PersonShiftAssignments	create	One period a person was assigned to work at a place. One record per person per shift.	Person	ONE_TO_MANY_OPTIONAL	subject	ingested	DRQ023
People	SBJ004	PersonSignatures	create	One signature a person applied to a record — what was signed, when, and what the signature meant. One record per signature.	Person	ONE_TO_MANY_OPTIONAL	occurrence	ingested	DRQ032
Processes	SBJ036	ProcessSegment	create	One step of work, defined independently of any product — mixing, pasteurising, filling. One record per step.	—	—	subject	mastered	DRQ017
Processes	SBJ042	ProcessParameterValues	create	One value set for a setting at one recipe level, for one product at one place. One record per recipe level per setting.	Process Recipe Levels	ONE_TO_MANY_OPTIONAL	subject	mastered	DRQ017 · DRQ018

Domain	Concept ID	Concept	Action	Definition	Parent	Cardinality to parent	Kind	Mastery	From
Processes	SBJ041	ProcessRecipe Levels	create	One statement of a routing at one level of specificity — general, site, master, or the instructions for one batch. One record per routing per level per scope.	Process Routings	ONE_TO_MANY_MANDATORY	subject	mastered	DRQ018
Processes	SBJ038	ProcessParameters	create	One setting a step can be run at, with the range allowed. One record per step per setting.	Process Segment	ONE_TO_MANY_OPTIONAL	subject	mastered	DRQ017
Processes	SBJ040	ProcessRoutings	create	The steps, in order, that turn a product's inputs into the product. One record per routing.	Process Segment	ONE_TO_MANY_OPTIONAL	subject	mastered	DRQ017
Processes	SBJ039	Process SamplingPlans	create	One rule stating what is sampled at a step, when, how much, and against which specification. One record per step per sampling plan.	Process Segment	ONE_TO_MANY_OPTIONAL	subject	mastered	DRQ025
Processes	SBJ037	Process Segment Requirements	create	One thing a step needs to run — a capability, a qualification, a kind of material. One record per step per requirement.	Process Segment	ONE_TO_MANY_OPTIONAL	subject	mastered	DRQ017
Products	SBJ012	Product	create	One thing the organisation buys, makes or sells, at one level of processing or packing. One record per product.	—	—	subject	mastered	DRQ001 · DRQ013
Products	SBJ016	ProductBatches	create	One quantity made or received as one unit of traceability, with its lot code and date. One record per product per lot code.	Product	ONE_TO_MANY_OPTIONAL	subject	ingested	DRQ012 · DRQ030
Products	SBJ024	ProductCostings	create	The expected or actual cost of a product, by element, for a period. One record per product per cost element per period.	Product	ONE_TO_MANY_OPTIONAL	subject	derived	DRQ036

Domain	Concept ID	Concept	Action	Definition	Parent	Cardinality to parent	Kind	Mastery	From
Products	SBJ019	ProductDisclosures	create	One ingredient a product must disclose, how much of it there is, and whether it is withheld as confidential. One record per product per disclosed ingredient.	Product	ONE_TO_MANY_OPTIONAL	subject	mastered	DRQ010
Products	SBJ018	ProductHazards	create	One hazard a product carries — an allergen, an irritant, a classified chemical — with the basis of the declaration. One record per product per hazard.	Product	ONE_TO_MANY_OPTIONAL	subject	mastered	DRQ009
Products	SBJ013	ProductIdentifiers	create	One identifier by which a product is known, in one scheme. One record per product per scheme per identifier.	Product	ONE_TO_MANY_OPTIONAL	subject	mastered	DRQ014 · DRQ035
Products	SBJ025	ProductIdentity Changes	create	One change to a product, and whether it made a new product or a new version of what goes in. One record per product per change.	Product	ONE_TO_MANY_OPTIONAL	occurrence	mastered	DRQ014
Products	SBJ020	ProductOrigins	create	One stated country or region of origin, with the basis of the statement. One record per product per origin claim.	Product	ONE_TO_MANY_OPTIONAL	subject	ingested	DRQ009
Products	SBJ021	ProductShelf Lives	create	How long a product lasts under one set of storage conditions. One record per product per storage condition.	Product	ONE_TO_MANY_OPTIONAL	subject	mastered	DRQ011
Products	SBJ014	ProductSpecifications	create	One version of what a product must be, with its limits and how conformance is checked. One record per product per specification version.	Product	ONE_TO_MANY_OPTIONAL	subject	mastered	DRQ025
Products	SBJ015	ProductStructureVersions	create	One complete statement of what goes into a product, for a site, market or period. One record per product per version.	Product	ONE_TO_MANY_OPTIONAL	subject	mastered	DRQ001 · DRQ015

Domain	Concept ID	Concept	Action	Definition	Parent	Cardinality to parent	Kind	Mastery	From
Products	SBJ022	ProductSubstitutions	create	One approved alternative for an input, with the conditions for using it. One record per product per approved alternative.	Product	ONE_TO_MANY_OPTIONAL	subject	mastered	DRQ005
Products	SBJ023	ProductVariant Axes	create	One way the members of a family differ — size, format, pack count — and the values allowed. One record per family per axis.	Product	ONE_TO_MANY_OPTIONAL	subject	mastered	DRQ014
Products	SBJ017	ProductSub Batches	create	One part of a batch separated for handling, retaining its link to the batch. One record per batch per part.	Product Batches	ONE_TO_MANY_OPTIONAL	subject	ingested	DRQ012
Transactions	SBJ046	Transaction	create	One exchange of goods of a stated kind — a purchase, a receipt, an issue to production, a transfer, a despatch. One record per exchange.	—	—	occurrence	ingested	DRQ024 · DRQ033
Transactions	SBJ048	Transaction InventoryPositions	create	One quantity of a product, in one condition, at one place, at one time. One record per product per condition per place per count.	Transaction	ONE_TO_MANY_OPTIONAL	subject	ingested	DRQ013
Transactions	SBJ047	Transaction Lines	create	One line of an exchange: one product, one quantity, one batch. One record per exchange per line.	Transaction	ONE_TO_MANY_MANDATORY	occurrence	ingested	DRQ024

5.4 Lookups and Values

LKP009 · UniversalUnitOfMeasure · universal

The unit a quantity is expressed in. Used wherever a quantity is recorded.

Used by: [ProductBillOfMaterialsUnitOfMeasure](#)

Code	Name	Definition
KG	Kilogram	Mass in kilograms.
G	Gram	Mass in grams.
L	Litre	Volume in litres.
ML	Millilitre	Volume in millilitres.
EA	Each	A count of units.
CASE	Case	A count of cases.

LKP005 · EquipmentPositionLevelType · concept

The level of a place in the plant structure, following ISA-95.

Used by: [EquipmentPositionLevel](#)

Code	Name	Definition
SITE	Site	One manufacturing location.
AREA	Area	A part of a site — processing, packing, storage.
PROCESSCELL	Process cell	A work centre for batch production.
PRODUCTIONLINE	Production line	A work centre for repetitive production.
STORAGEZONE	Storage zone	A work centre for holding material.
WORKUNIT	Work unit	One part of a work centre performing one function.
EQUIPMENTMODULE	Equipment module	One part of a unit performing one task.

LKP007 · EventDispositionsDecisionType · concept

What is done with material that failed its specification.

Used by: [EventDispositionsDecision](#)

Code	Name	Definition
USEASIS	Use as is	Accepted despite the failure, on a stated justification.
REWORK	Rework	Returned to production to be brought into specification.
REGRADE	Regrade	Accepted against a different specification.
SCRAP	Scrap	Destroyed or disposed of.
RETURN	Return to supplier	Sent back to the party that supplied it.

LKP006 · EventStoppagesReasonType · concept

Why production stopped, from a governed list.

Used by: [EventStoppagesReason](#)

Code	Name	Definition
CHANGEOVER	Changeover	Moving the line between products.
CLEANING	Cleaning	Scheduled cleaning or sanitation.
PLANNEDMAINT	Planned maintenance	Work scheduled against a plan.
BREAKDOWN	Breakdown	A machine ceased to perform its function.
MATERIALSHORT	Material shortage	An input was unavailable.
QUALITYHOLD	Quality hold	Stopped pending a quality decision.
UPSTREAM	Upstream blockage	Starved by a preceding place.
DOWNSTREAM	Downstream blockage	Blocked by a following place.
LABOUR	Labour unavailable	No qualified operator available.
UTILITY	Utility failure	Power, water, steam or air unavailable.

LKP004 · ProcessRecipeLevelsType · concept

How specific a recipe is, following ISA-88.

Used by: [ProcessRecipeLevelsLevel](#)

Code	Name	Definition
GENERAL	General recipe	The formulation and method, naming no factory.
SITE	Site recipe	Adapted to the materials and constraints of one site.
MASTER	Master recipe	Bound to one work centre, with its settings.
CONTROL	Control recipe	The instructions issued for one batch. This is what runs.

LKP002 · ProductHazardType · concept

The kind of hazard a product carries, and therefore which declaration regime applies to it.

Used by: [ProductHazardsType](#)

Code	Name	Definition
FOODALLERGEN	Food allergen	One of the major food allergens, declared by the food it comes from.
SENSITISER	Skin sensitiser	A substance that may cause a reaction on contact.
CLASSIFIEDCHEMICAL	Classified chemical	A substance carrying a hazard classification requiring disclosure.
PHYSICAL	Physical hazard	A foreign body or sharp-edge risk.

LKP003 · ProductIdentifiersType · concept

The scheme an identifier belongs to.

Used by: [ProductIdentifiersType](#)

Code	Name	Definition
GTIN	Trade item number	The global trade item number allocated under GS1 rules.
MATERIALNUMBER	Internal material number	The organisation's own number for the product.
SUPPLIERPART	Supplier part number	The supplier's number for what it sells.

LKP001 · ProductKindType · concept

What a product is to this organisation, deciding whether it may be bought, made, used or sold. A product may be more than one kind at once.

Used by: [ProductKind](#)

Code	Name	Definition
RAWMATERIAL	Raw material	An ingredient or substance that enters a formulation.
PACKCOMPONENT	Packaging component	A container, closure, label, case or divider.
INDIRECT	Indirect material	Used in production without entering the product.
INTERMEDIATE	Part-made good	Made and used internally; neither bought nor sold.
CONSUMERUNIT	Consumer unit	The unit a consumer buys.
CASE	Case	The unit a customer orders.
PALLET	Pallet	The unit that ships.
FAMILY	Family	Not a physical thing: the group a set of variants belongs to.

LKP008 · TransactionKindType · concept

What kind of exchange a transaction is.

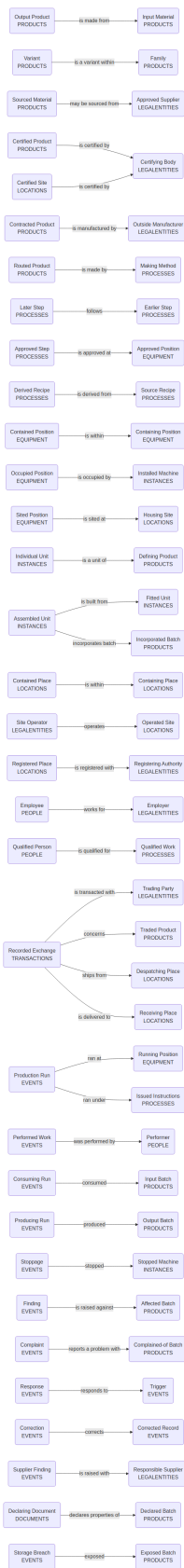
Used by: [TransactionKind](#)

Code	Name	Definition
PURCHASE	Purchase	An order placed with a supplier.
RECEIPT	Receipt	Goods arriving and becoming the organisation's.
ISSUE	Issue to production	Material released to a run.
TRANSFER	Transfer	A move between the organisation's own places.
DESPATCH	Despatch	Goods leaving for a customer.

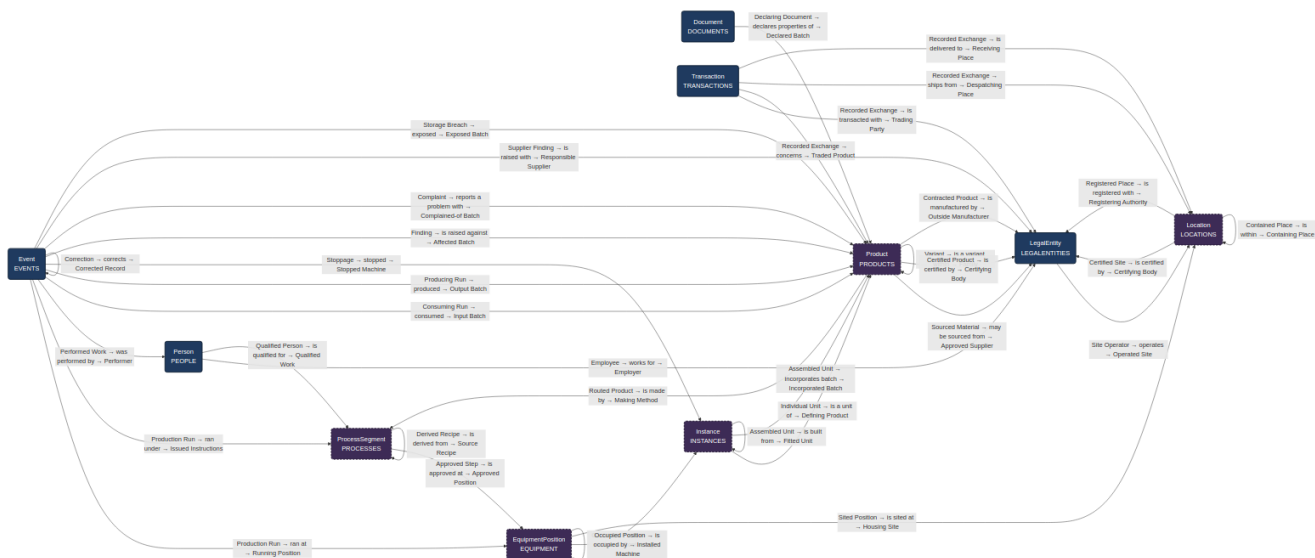
5.5 Diagrams

Every diagram is drawn the same way: a box names the thing on its first line and qualifies it on the second, and an edge carries the verb.

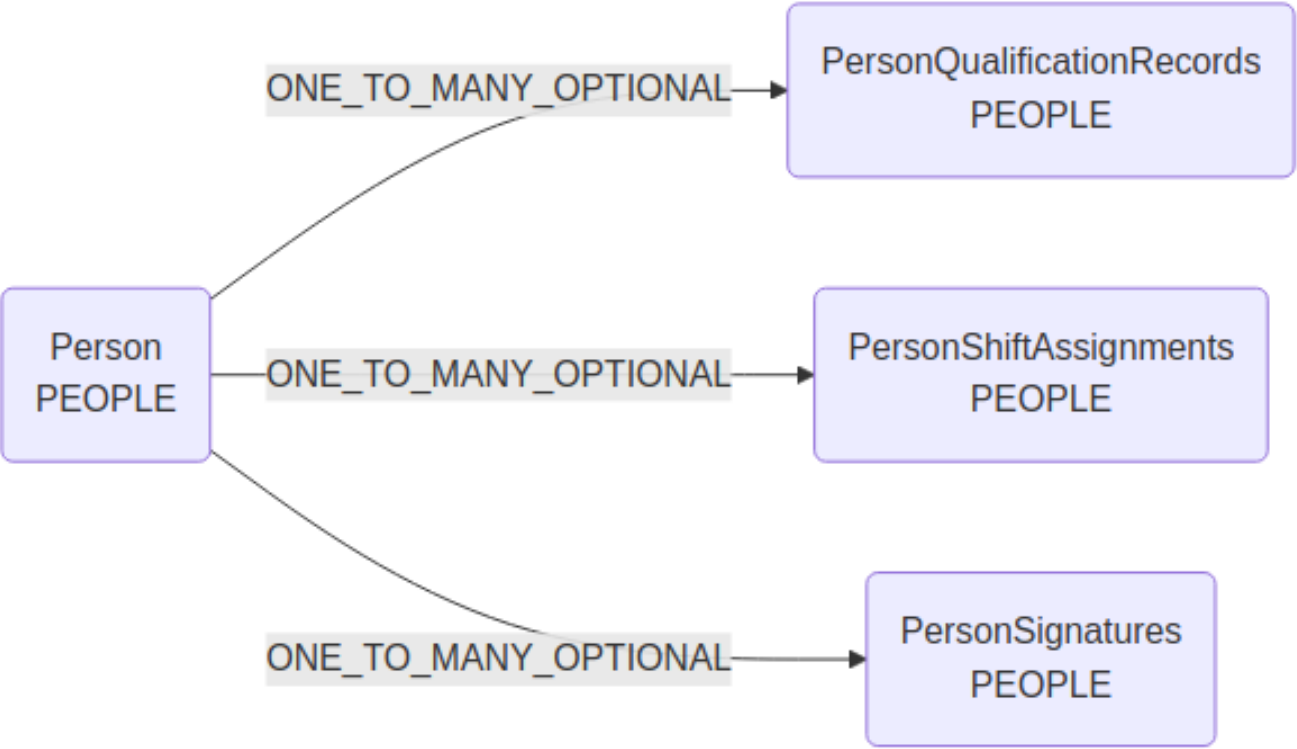
Roles data model. One node per role, one edge per relationship. A role nothing connects to is a gap.



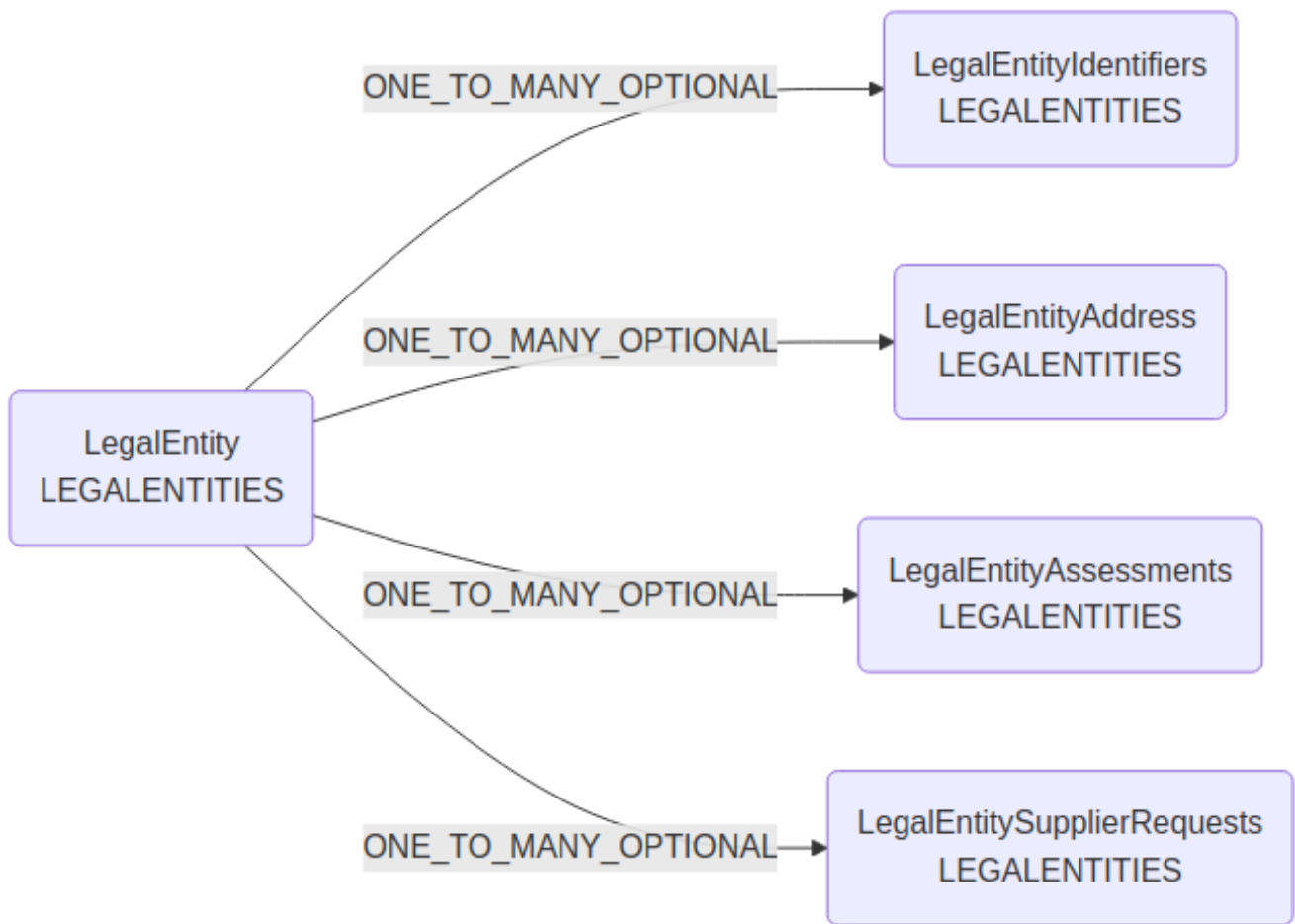
Domains data model. The same relationships seen from the domains, each edge carrying the roles it confers. Domains DDA holds are solid; proposed domains are dashed.



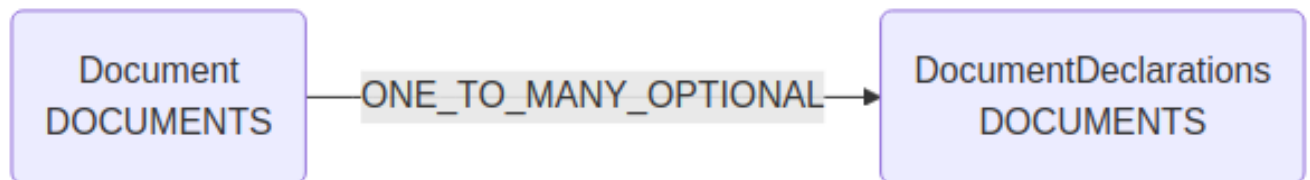
People concepts.



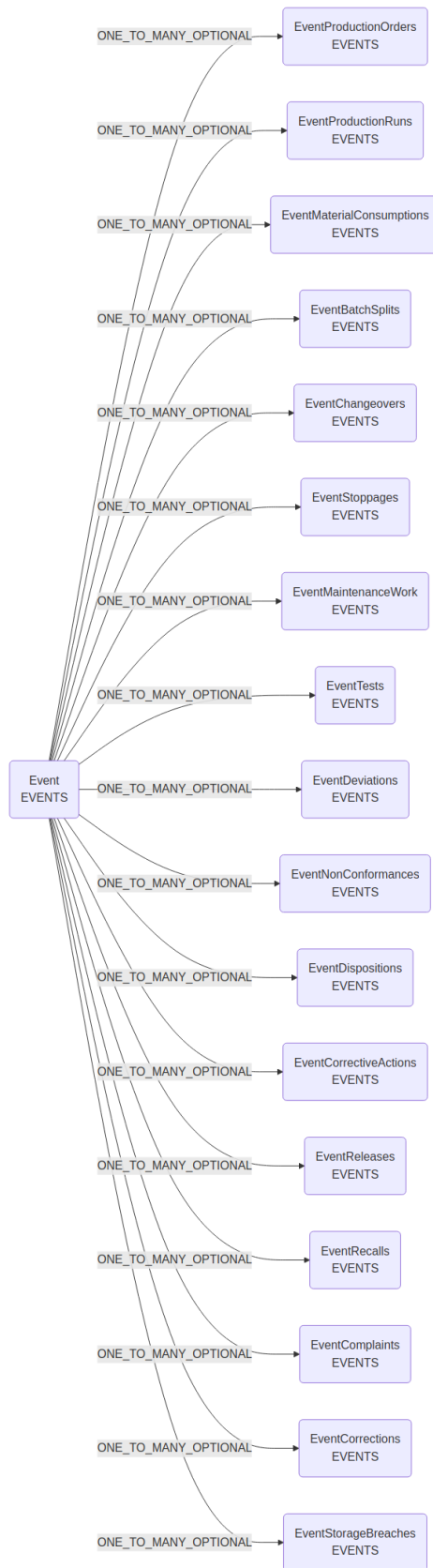
LegalEntities concepts.



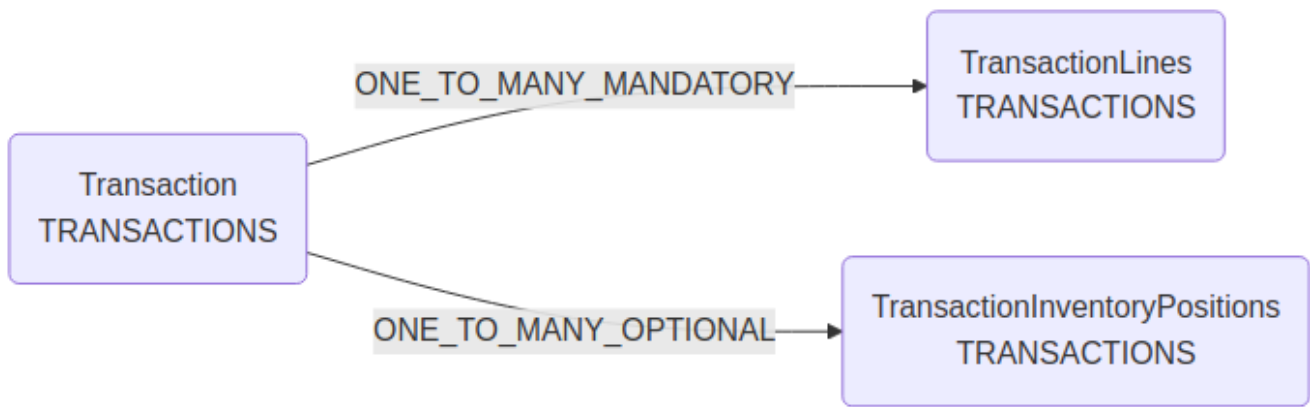
Documents concepts.



Events concepts.



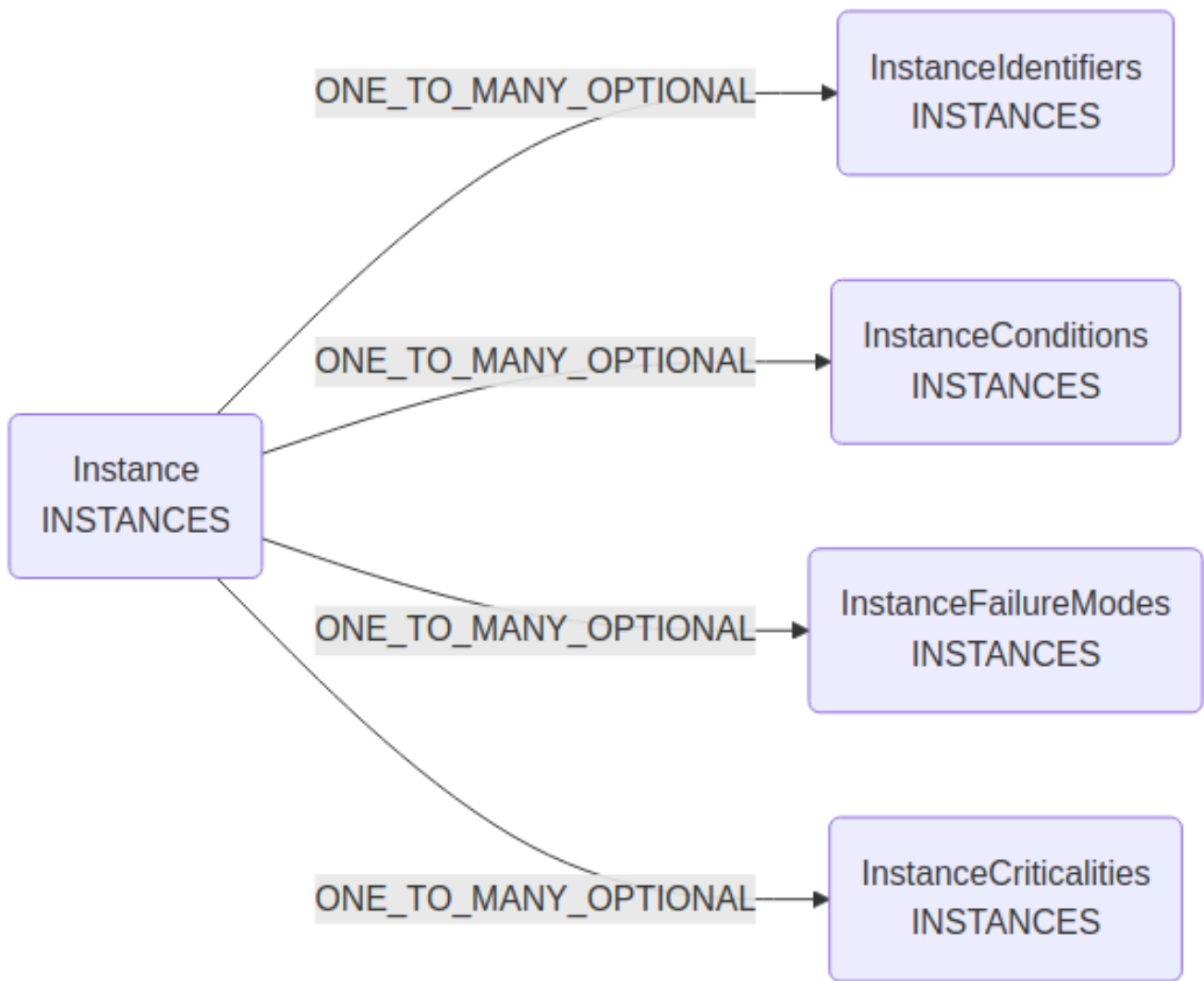
Transactions concepts.



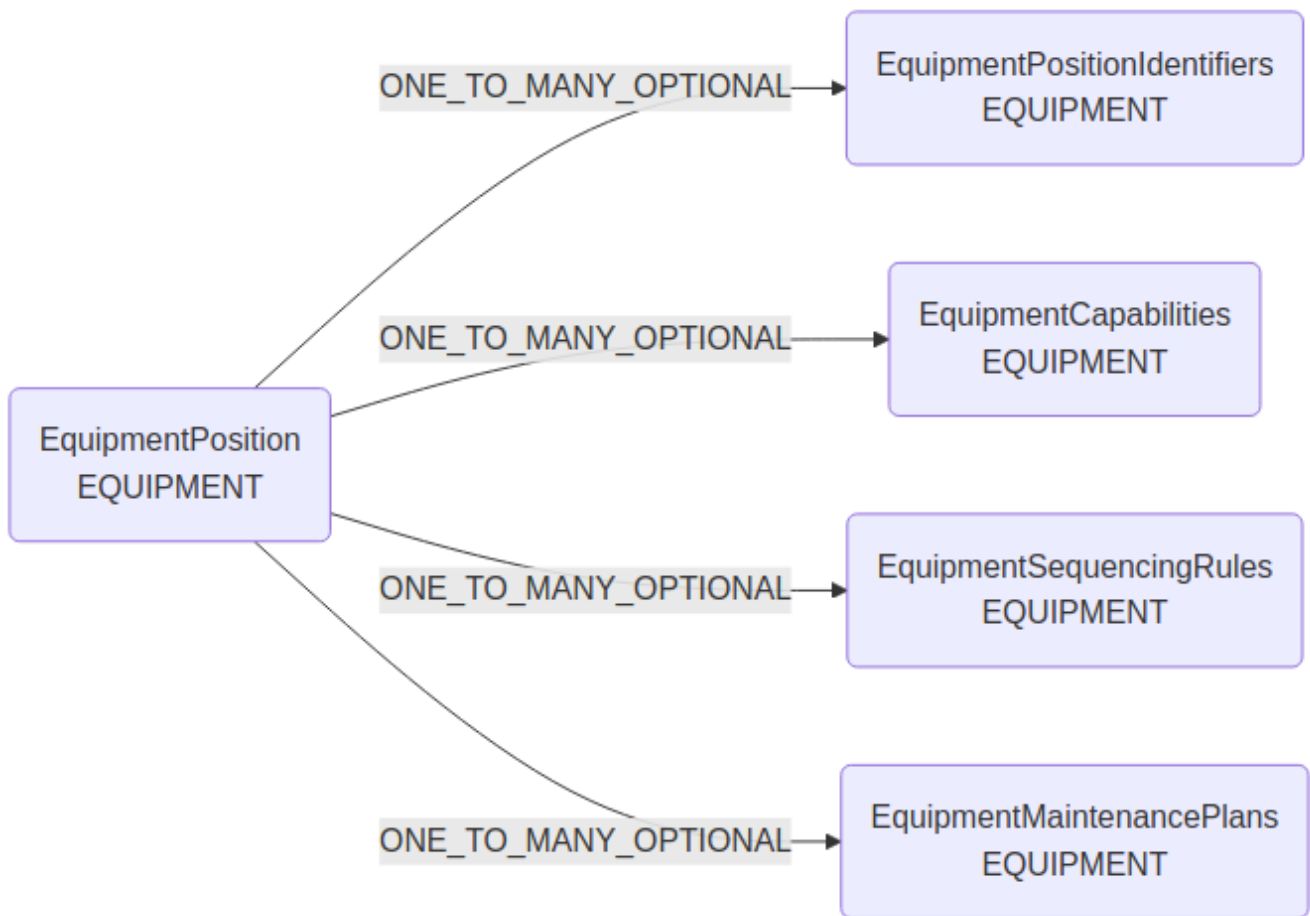
Products concepts.



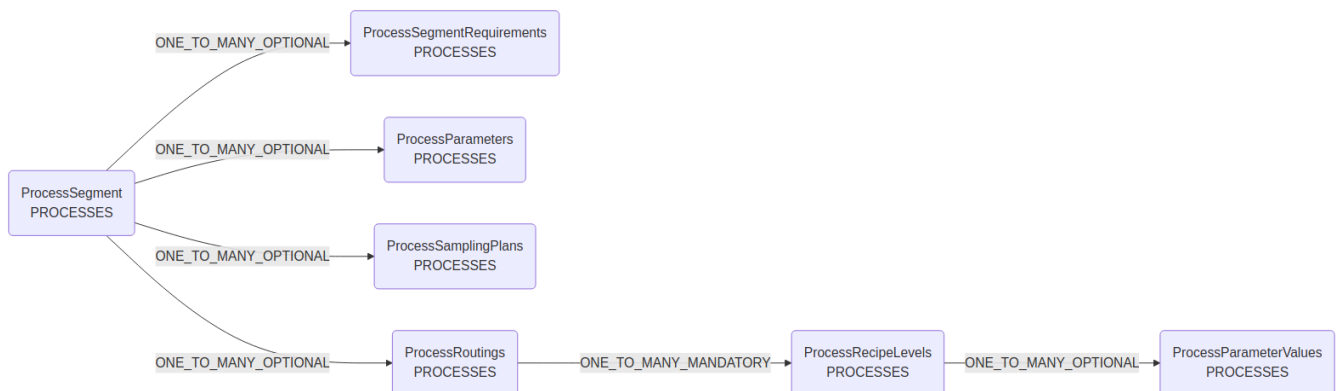
Instances concepts.



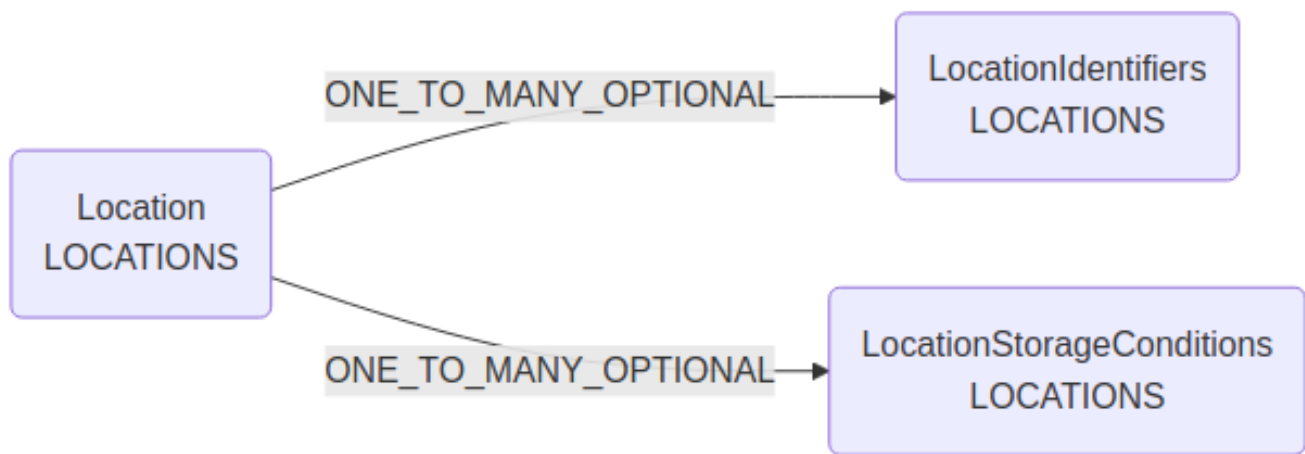
Equipment concepts.



Processes concepts.



Locations concepts.



6 Source Validation

Source	Establishes	Answered by
SRC001	ISA-95 sets out a role-based equipment hierarchy — enterprise, site, area, work centre, work unit, equipment module and control module — and records a person's qualifications alongside results.	REQ026 · REQ033 · REQ034 · REQ041 · REQ064 · REQ065 · SBJ001 · SBJ002 · SBJ031 · SBJ032 · SBJ033 · SBJ037 · REL010 · REL012 · REL021 · ATR054 · ATR055 · ATR074 · ATR075 · LKP005 · ROL020 · ROL021 · ROL022 · ROL024 · ROL025 · ROL039 · ROL040
SRC002	A batch procedure is an ordered set of unit procedures, operations and phases; recipes are general, site, master or control, and carry a formula, equipment requirements and a procedure.	REQ002 · REQ021 · REQ025 · REQ026 · REQ029 · SBJ015 · SBJ036 · SBJ037 · SBJ040 · SBJ041 · REL006 · REL007 · REL009 · ATR004 · ATR026 · ATR027 · ATR028 · ATR060 · ATR061 · ATR065 · ATR068 · LKP004 · ROL012 · ROL013 · ROL014 · ROL015 · ROL018 · ROL019
SRC003	S88 separates what is to be done from how equipment is driven, with parameter limits, recipe hierarchy, and batch records under electronic-record rules.	REQ025 · REQ027 · REQ028 · REQ029 · REQ030 · REQ050 · SBJ038 · SBJ040 · SBJ041 · SBJ042 · SBJ062 · REL008 · REL027 · ATR062 · ATR063 · ATR066 · ATR116 · LKP004 · ROL016 · ROL017 · ROL019 · ROL048
SRC004	The equipment model says what the plant can do and the recipe says what to do; a recipe asks for 'heat to 80 for 20 minutes' and a new product variant becomes a new recipe, not new code.	REQ027 · REQ028 · REQ033 · REQ034 · SBJ033 · SBJ036 · SBJ038 · SBJ042 · REL011 · ATR056 · ATR064 · ROL013 · ROL017 · ROL023
SRC005	Genealogy links every raw-material lot, intermediate, step, test, packaging operation and shipping unit; the bill of materials defines expected links and the batch record the links that actually occurred, including splits and merges.	REQ001 · REQ002 · REQ003 · REQ004 · REQ005 · REQ006 · REQ007 · REQ016 · REQ017 · REQ022 · REQ040 · REQ050 · REQ054 · REQ068 · SBJ012 · SBJ015 · SBJ016 · SBJ017 · SBJ022 · SBJ049 · SBJ051 · SBJ052 · SBJ053 · SBJ062 · REL001 · REL029 · REL030 · ATR001 · ATR002 · ATR003 · ATR005 · ATR011 · ATR012 · ATR017 · ATR032 · ATR033 · ATR042 · ATR095 · ATR096 · ATR115 · LKP001 · LKP009 · ROL001 · ROL002 · ROL051 · ROL052 · ROL053
SRC006	Manufacturers need to answer which raw-material lots went into a batch, which finished products were affected, which operators and equipment were involved, and whether checks were completed.	REQ003 · REQ005 · REQ041 · ROL050
SRC007	Many manufacturers track some parts by serial number and bulk materials by batch; a blend of two lots must record both; returned stock keeps its batch identity and remaining shelf life.	REQ011 · REQ015 · REQ016 · REQ023 · SBJ017 · SBJ021 · SBJ026 · SBJ053 · REL014 · REL015 · ATR006 · ATR031 · ATR041 · ATR048 · ROL026 · ROL028 · ROL029 · ROL030
SRC008	Food and beverage production systems manage recipes, formulation, lot traceability and allergen control.	REQ002 · REQ012

Source	Establishes	Answered by
SRC009	Businesses handling listed foods must keep key data elements for critical tracking events and provide them to the FDA within 24 hours, and must share information with supply-chain partners.	REQ042 · REQ051 · REQ053 · REQ063 · SBJ063 · REL024 · REL025 · ATR086 · ATR117 · ROL044 · ROL045 · ROL052
SRC010	The FDA has proposed extending the FSMA 204 compliance date to 20 July 2028; critical tracking events include packing, shipping and receiving.	REQ051 · SBJ063
SRC011	A traceability lot code is assigned at initial packing or transformation, linked at each tracking event, exchanged between trading partners, and kept for two years; its source may be identified by registration number, location number or web address.	REQ052 · REQ053 · REQ063 · REQ067 · REQ069 · SBJ006 · SBJ016 · SBJ043 · SBJ044 · REL018 · ATR029 · ATR030 · ATR070 · ATR071 · ROL035 · ROL036 · ROL054
SRC012	A deviation is a departure from an approved procedure; a nonconformance is a failure to meet a requirement; nonconformances are classified minor, major or critical.	REQ044 · REQ046 · SBJ014 · SBJ058 · SBJ059 · ATR106 · ATR108 · ROL057
SRC013	Deviations are recorded, risk-classified and investigated, lead to corrective action, and support decisions about what to do with the batch.	REQ046 · REQ047 · REQ048 · REQ049 · SBJ058 · SBJ061 · REL034 · ATR112 · ROL061 · ROL062
SRC014	Affected material is contained, a disposition decided — use as is, rework, repair, scrap or return to supplier — with approvals, root cause found for significant or repeat issues, and closure verified.	REQ018 · REQ047 · REQ049 · REQ059 · REQ070 · SBJ009 · SBJ024 · SBJ059 · SBJ060 · SBJ061 · REL032 · REL036 · ATR044 · ATR045 · ATR082 · ATR107 · ATR109 · ATR110 · ATR111 · ATR113 · ATR114 · LKP007 · ROL057 · ROL058 · ROL065 · ROL066
SRC015	US GMP rule 21 CFR 211.192 requires a thorough investigation of any deviation, with conclusions and follow-up recorded.	REQ048
SRC016	The production system often knows a line stopped with a reason code while the maintenance system still shows the machine as available; connecting them makes one stoppage drive both.	REQ035 · REQ037 · REQ039 · SBJ035 · SBJ055 · SBJ056 · REL026 · REL031 · ATR009 · ATR100 · ATR101 · ATR102 · ROL047 · ROL055 · ROL056
SRC017	Downtime reasons must be standardised into tiers; when operators pick from many vague reasons, a changeover is logged four different ways.	REQ032 · REQ038 · SBJ054 · SBJ055 · ATR097 · ATR099 · LKP006
SRC018	Equipment effectiveness is availability times performance times quality; food plants start with structured failure codes on every stoppage and focus on their most critical machines.	REQ035 · REQ036 · REQ038 · REQ039 · REQ070 · SBJ028 · SBJ029 · SBJ030 · SBJ035 · SBJ056 · ATR013 · ATR050 · ATR051 · ATR052 · ATR053 · ATR059 · LKP006 · ROL023 · ROL056

Source	Establishes	Answered by
SRC019	ISO 22716 requires documentation of every stage of cosmetic production from raw materials to distribution, and US cosmetics law now aligns with it on traceability.	REQ014
SRC020	Raw materials require supplier qualification, testing against specification, documented receipt, controlled storage, and batch tracking with deviation management.	REQ008 · REQ009 · REQ044 · REQ045 · REQ058 · REQ059 · REQ066 · SBJ005 · SBJ007 · SBJ008 · SBJ009 · SBJ011 · SBJ014 · SBJ039 · SBJ045 · SBJ057 · SBJ066 · REL003 · REL038 · ATR007 · ATR014 · ATR015 · ATR023 · ATR024 · ATR025 · ATR067 · ATR072 · ATR073 · ATR080 · ATR081 · ATR103 · ATR104 · ATR105 · ATR120 · ROL005 · ROL006 · ROL066 · ROL068 · ROL069 · ROL070
SRC021	Documentation should capture every aspect of manufacturing so that problems can be traced and corrected.	REQ040
SRC022	ISO 22716 covers personnel, premises, equipment and maintenance, materials, production, quality control, contract and outsourced activities, recalls and complaints, and expects any batch to be reconstructable.	REQ045 · REQ054 · REQ062 · REQ064 · SBJ002 · SBJ057 · SBJ064 · REL017 · REL019 · REL033 · ATR118 · ROL008 · ROL009 · ROL033 · ROL034 · ROL039 · ROL059 · ROL060
SRC023	In tolling the brand supplies its own raw materials; in turnkey manufacturing the contract manufacturer provides them.	REQ010 · REQ060 · REQ061 · REL022 · ATR087 · ROL011 · ROL042
SRC024	In turnkey contract manufacturing the manufacturer sources materials, makes, checks and packs the product; the client provides specifications and approvals.	REQ010 · REQ060 · REQ061 · REL005 · ATR008 · ATR018 · ATR036 · ROL010 · ROL011
SRC025	Household cleaning manufacturers must disclose each ingredient, its CAS number, its function and a link to its safety data sheet; some may be withheld as confidential under a generic name.	REQ013 · REQ014 · SBJ019 · ATR037 · ATR039 · LKP002
SRC026	Manufacturers of household cleansing products furnish each ingredient and its content by weight; confidential business information may be withheld.	REQ013 · SBJ019 · ATR038
SRC027	Facilities making animal food, including pet food, must comply with good manufacturing practice under 21 CFR Part 507.	REQ040
SRC028	Pet food uses more than 500 ingredients; manufacturers identify hazards and put preventive controls such as heat treatment in place; the FDA regulates both finished pet food and its ingredients.	REQ014

Source	Establishes	Answered by
SRC029	Responsible pet food makers run supplier approval, precise formulation, and batch records of finished product and where it went.	REQ008 · REQ058 · SBJ008 · ROL006
SRC030	A change to declared formulation — such as adding nuts, which introduces an allergen — net content, a certification mark or pack quantity requires a new product number at every packing level.	REQ004 · REQ011 · REQ019 · REQ020 · REQ021 · REQ069 · SBJ012 · SBJ013 · SBJ020 · SBJ023 · SBJ025 · REL002 · REL004 · ATR016 · ATR019 · ATR020 · ATR021 · ATR022 · ATR040 · ATR043 · ATR046 · LKP001 · LKP003 · ROL003 · ROL007 · ROL009
SRC031	Formulation is the list of ingredients or components used to create a trade item; any change to the declared net content requires a new product number.	REQ019 · REQ020 · SBJ013 · SBJ023 · SBJ025 · ATR047 · LKP003 · ROL004
SRC032	Part 11 sets when electronic records and signatures are acceptable — identity, authority, auditability and retention — across food, cosmetics and other regulated manufacturing.	REQ055 · REQ056 · SBJ004 · SBJ065 · REL035 · ATR077 · ATR078 · ATR119 · ROL063 · ROL064
SRC033	A batch record documents each step — by whom or by which equipment, when, with what materials and quantities, and with what results — including checks by a second qualified person.	REQ006 · REQ007 · REQ022 · REQ030 · REQ040 · REQ041 · REQ055 · REQ057 · SBJ003 · SBJ004 · SBJ022 · SBJ050 · SBJ051 · SBJ052 · REL020 · REL028 · ATR010 · ATR076 · ATR079 · ATR092 · ATR093 · ATR094 · ROL037 · ROL038 · ROL046 · ROL049 · ROL050
SRC034	Allergens must be declared; cross-contact arises from rework, carry-over on shared equipment and production sequencing.	REQ012 · REQ031 · REQ032 · SBJ018 · SBJ034 · SBJ054 · ATR057 · ATR058 · ATR098
SRC035	Nine major food allergens must be declared by the food they come from — whey must be declared as milk.	REQ011 · REQ012 · SBJ018 · ATR034 · ATR035 · LKP002
SRC036	Food allergen labelling rules do not apply to pet foods, animal feeds, cosmetics, drugs or household cleaning products.	REQ014
SRC037	A GS1 2D code carries the product number, batch, production date, expiry date and serial number; retailers are to accept 2D codes by the end of 2027.	REQ023 · REQ024 · SBJ026 · SBJ027 · REL013 · ATR049 · ROL026 · ROL027
SRC038	Sunrise 2027 is not a legal requirement, but major retailers are aligning to it, with packs carrying both codes during the transition.	REQ024
SRC039	At receipt, capture supplier lot, quantity, expiry and certificate reference; record exact storage locations; issue lot-tracked materials to production orders.	REQ001 · REQ009 · REQ015 · REQ017 · REQ065 · SBJ010 · SBJ011 · SBJ021 · SBJ043 · SBJ046 · SBJ047 · SBJ048 · REL016 · REL023 · REL037 · ATR069 · ATR083 · ATR084 · ATR088 · ATR090 · ATR091 · LKP008 · ROL031 · ROL032 · ROL043 · ROL067

Source	Establishes	Answered by
SRC040	Traceability links identity (product number, lot, serial), status (quarantined, released) and events (receipt, consume, transform, pack, ship, return).	REQ018 · REQ042 · SBJ046 · SBJ048 · ATR085 · ATR089 · LKP008 · ROL041

Coverage of the scope.

ID	In scope	Sources
SCI001	Buying materials, and deciding which suppliers may supply them	4
SCI002	Defining products — what they are made from and what they must be	10
SCI003	Defining how products are made — the steps, the settings and the recipes	3
SCI004	The layout of each factory and the machines in it	5
SCI005	Making product, and what each batch actually used and produced	9
SCI006	Testing, recording problems, putting them right, and releasing product	8
SCI007	Tracing any batch back to its ingredients and forward to where it went	11
SCI008	Stock held at the organisation's own sites	3
SCI009	Product made or packed by outside manufacturers	3

7 Existing Model Findings

ID	DDA record	Check	Finding
PLF001	DMC000000011	M01 / DMS000000169 principle 1	LegalEntityRoles stores roles — investor, vendor, client — on the organisation. Roles come from the relationships in force; this model derives supplier, outside manufacturer and certifying body instead.
PLF002	DMC000000024	M01, naming	PeopleRoles stores roles on the person, and uses the plural People where its siblings use the singular Person.
PLF003	DMC000000107	M10	LegalEntityRelationships holds relationships as a concept. DDA now holds relationships as their own records; not reused here.
PLF004	DMD000000020	DT-anchor	The Events domain holds no concepts, so it has no anchor. Its definition describes corporate and life events only, and does not cover operational occurrences.
PLF005	DMD000000022	DT-anchor	The Transactions domain holds no concepts, so it has no anchor. Its definition is written for trades in securities and does not cover exchanges of goods.
PLF006	DMC000000142	naming	The Document concept's description calls it DocumentBase, which disagrees with its name.
PLF007	DMC000000147	naming	PersonCitizenship is singular where its siblings are plural.
PLF008	DMR000000001	R01	The relationship reads Managed By forward and Invests In back. The inverse does not read the forward verb back, so the pair describes two arrangements under one name.
PLF009	DMR000000002	R01	The forward verb is Investor, which is a noun. A role has been entered in the verb field.
PLF010	DMR000000001	R06	The stored forward view is LegalEntityGP; the roles on the record derive Legal EntityGPs. A stored derived name has drifted from what it derives from.

Guide	Records	Finding
dda://guides/data-design-authority	list_model(state Transitions)	The guide lists state transitions as readable; the call returns an internal error, so the change package carries no preconditions.

8 Conformance to DDA Standards

Standard	Name	Category	Applies	How satisfied
DMS000000141	Data is a shared, valuable enterprise asset	Data Principles	yes	Every domain in the model is an enterprise subject referenced by more than one operating model; section 5.1 records which.
DMS000000142	Everyone has a role in good data management	Data Principles	yes	Each requirement names an accountable owner, and each finding names who it belongs to.
DMS000000143	The Chief Data Officer owns and approves the Data Governance Policy	Policy Governance	no	Policy ownership. Nothing in this model changes who owns the policy.
DMS000000144	Data Governance Policy aligns with and defers to more specific policies	Policy Governance	no	Policy precedence. No conflict with a more specific policy arises in this model.
DMS000000145	Data Governance Policy applies to all critical business data	Policy Scope	no	Policy scope. The model does not decide which data the policy applies to.
DMS000000146	A Data Governance Committee must be established	Governance Structure	yes	Five domains are proposed to the Committee, each with its ten domain tests and the domains considered instead.
DMS000000147	Data domains must be defined based on the subject of the data	Data Domains	yes	Every domain is named for the subject its records are about. Eight activity- and role-shaped candidates were rejected and are recorded with the test each failed.
DMS000000148	Standard data governance roles must be defined	Roles and Responsibilities	yes	Each domain proposal names the obligations that would give it its own steward.
DMS000000149	Each data domain must have a single authoritative data model	Data Modeling	yes	Each subject sits in one domain only, and each domain has exactly one anchor concept.

Standard	Name	Category	Applies	How satisfied
DMS000000150	Each data domain must have an Authoritative Data Repository	Authoritative Sources	no	Authoritative repositories are named per domain by the Committee, not by a model design.
DMS000000151	Primary record identifiers must be established by domain	Data Identification	yes	Each proposed domain records the identifier scheme by which its records are uniquely known, under test DT7.
DMS000000152	Critical Data Elements must be identified and governed	Data Quality	yes	Attributes supporting regulatory, safety or financial obligations are marked critical.
DMS000000153	A Data Issues for Management Resolution program must be established	Issue Management	no	Issue management is a programme, not a model artefact.
DMS000000154	Changes to the Data Governance Policy must be formally controlled	Policy Lifecycle	yes	The package carries its method version, and changes to it are governed by the method rather than by this engagement.
DMS000000168	Data lineage must be defined and maintained for all critical data	Data Lineage	yes	Lineage is held by the platform and is not modelled; batch-to-batch lineage in the business sense is held as relationships between runs and batches.
DMS000000169	Data Model Design Principles	APPENDIX	yes	The design states how it meets each of the fourteen principles: subjects modelled once with roles derived from relationships (1, 7); business meaning approved before structure (2, 9, 14); sources understood before integration (3); enterprise identity established per domain (4); the model published rather than defined in reports (5, 6, 10); the model generating implementation through the change package (8); sensitivity, criticality and retention declared on attributes (13). Principles 11 and 12 concern data products and are met at product design rather than here.
DMS000000156	Definition of Data Domain	Glossary	yes	The glossary standards define domain, model, concept, attribute, producer, consumer, custodian, operational data store, master data management, warehouse, lake and critical data element. The model uses each term as defined. Covers: DMS000000157, DMS000000158, DMS000000159, DMS000000160, DMS000000161, DMS000000162, DMS000000163, DMS000000164, DMS000000165, DMS000000166, DMS000000167

9 Exceptions

ID	Standard	Applies to	Justification	Consequence	Owner	Route back	Approved
EXC001	M04	SBJ012	Product anchors physical things and also the family groupings that variants belong to, which are not physical.	A reader querying products by kind must exclude families, or count groupings as goods.	Technical lead	A separate grouping domain if families acquire obligations of their own.	Not yet
EXC002	M15	REL015 · REL029 · REL030 · REL032 · REL033 · REL037 · REL038	These relationships point at a product batch, which sits beneath the Product anchor rather than being the anchor itself. Traceability is between batches, not between product definitions.	A relationship's destination must be read as a concept within the destination domain rather than its anchor.	Quality manager	A declared sub-anchor destination in the method, or Batches admitted as a domain.	Not yet

10 Outstanding Decisions

ID	Decision	Options
DEC001	Approve EXC001 and EXC002, or route them back.	approve · route back
DEC002	Certification of the requirement set cannot pass in research-only mode. Publish as a reference model, or appoint a business owner.	publish as reference model · appoint an owner
DEC003	Five domains and two anchor concepts in existing domains need the Data Governance Committee.	approve · decline
DEC004	Instances takes the prefix UNT because INS is held by Instruments.	confirm UNT · reassign
DEC005	Is finished output identified individually, or only by batch? (SCQ001)	individually · batch only · both
DEC006	The total cost of poor quality is asked for and no artefact answers it. Bring it into scope, or state it as out of scope.	in scope · out of scope
DEC007	State transitions could not be read: list_model(stateTransitions) returned "An internal error occurred". The change package therefore carries no preconditions.	retry before applying · apply without preconditions
DEC008	The Events and Transactions domain definitions in DDA are written for financial services. Propose broadening them.	propose update · leave as is

11 Approval

Block	Certifies	Status
Business requirements	The requirement set is complete and correct	Not certified — research-only, see DEC002
Data requirements	Accepted for implementation	Awaiting the two exceptions in section 9
Data Governance Committee	Five domains and two anchor concepts	Awaiting, see Appendix D and DEC003

The supplied sections — About This Document, 1.1 and 1.2 — and the practice's method rules in Appendix C come from `boilerplate.json`, which this run does not carry.

Appendix A — Sources

ID	Title	Organisation	Type	Why included	Establishes	Cited by
SRC001	OPC 10030 ISA-95 Common Object Model — ISA-95 Base Information Model	OPC Foundation	standard	The formal standard for how a factory is structured. The only source defining the levels of the plant, and the basis for keeping a place on a line separate from the machine in it.	ISA-95 sets out a role-based equipment hierarchy — enterprise, site, area, work centre, work unit, equipment module and control module — and records a person's qualifications alongside results.	27
SRC002	ISA-88	Wikipedia	reference-model	The standard for how batch recipes are structured. It establishes the four levels of a recipe and that a recipe names the equipment it needs.	A batch procedure is an ordered set of unit procedures, operations and phases; recipes are general, site, master or control, and carry a formula, equipment requirements and a procedure.	28
SRC003	ISA-88 (S88) Batch Control Standard	SG Systems Global	practitioner	A practitioner reading of ISA-88. It confirms that recipes and equipment are held apart in practice and that parameters carry limits.	S88 separates what is to be done from how equipment is driven, with parameter limits, recipe hierarchy, and batch records under electronic-record rules.	22
SRC004	ISA-88	Fabrico	practitioner	Shows the recipe and the method used across several products on one tank, and that the plant structure maps onto how maintenance records machines.	The equipment model says what the plant can do and the recipe says what to do; a recipe asks for 'heat to 80 for 20 minutes' and a new product variant becomes a new recipe, not new code.	14
SRC005	Batch Genealogy	SG Systems Global	practitioner	The clearest source separating what a batch should have used from what it did use. Also the main source on splitting and merging lots.	Genealogy links every raw-material lot, intermediate, step, test, packaging operation and shipping unit; the bill of materials defines expected links and the batch record the links that actually occurred, including splits and merges.	47

ID	Title	Organisation	Type	Why included	Establishes	Cited by
SRC006	MES Traceability and Genealogy	GE Vernova	practitioner	States the questions a plant is asked in plain terms. Several data requirements are those questions.	Manufacturers need to answer which raw-material lots went into a batch, which finished products were affected, which operators and equipment were involved, and whether checks were completed.	4
SRC007	How to do batch tracking for manufacturing	Cleverence	practitioner	The only source stating that serial and batch tracking are mixed within one operation.	Many manufacturers track some parts by serial number and bulk materials by batch; a blend of two lots must record both; returned stock keeps its batch identity and remaining shelf life.	18
SRC008	What is a manufacturing execution system?	IBM	reference-model	A general reference placing recipes, formulation and allergen control together in food manufacturing.	Food and beverage production systems manage recipes, formulation, lot traceability and allergen control.	2
SRC009	FSMA Final Rule on Requirements for Additional Traceability Records for Certain Foods	US Food and Drug Administration	regulatory	The regulation setting the twenty-four-hour limit on traceability records. The primary legal source for recall.	Businesses handling listed foods must keep key data elements for critical tracking events and provide them to the FDA within 24 hours, and must share information with supply-chain partners.	12
SRC010	FDA FSMA 204 Food Traceability Rule	NC State Extension	regulatory	Gives the current compliance date, which the model must be ready for.	The FDA has proposed extending the FSMA 204 compliance date to 20 July 2028; critical tracking events include packing, shipping and receiving.	2
SRC011	FSMA 204 General Summary	Famous Software	practitioner	States when lot codes are assigned, how long they are kept, and that places are identified by several authorities.	A traceability lot code is assigned at initial packing or transformation, linked at each tracking event, exchanged between trading partners, and kept for two years; its source may be identified by registration number, location number or web address.	17

ID	Title	Organisation	Type	Why included	Establishes	Cited by
SRC012	Nonconformance: Definition, Types, Causes, and Process	SimplerQMS	practitioner	Draws the line between a departure from procedure and a failure against specification.	A deviation is a departure from an approved procedure; a nonconformance is a failure to meet a requirement; nonconformances are classified minor, major or critical.	8
SRC013	Deviation Management Process in GMP	GMP Insiders	practitioner	Connects a departure from procedure to investigation, corrective action and the decision on the batch.	Deviations are recorded, risk-classified and investigated, lead to corrective action, and support decisions about what to do with the batch.	10
SRC014	Non-conformance report	Fabrico	practitioner	Sets out the disposition choices and that each needs approval.	Affected material is contained, a disposition decided — use as is, rework, repair, scrap or return to supplier — with approvals, root cause found for significant or repeat issues, and closure verified.	26
SRC015	Deviation management	Pharmaceutical Technology	practitioner	The regulatory basis for investigating every departure from procedure.	US GMP rule 21 CFR 211.192 requires a thorough investigation of any deviation, with conclusions and follow-up recorded.	1
SRC016	MES and CMMS integration	OxMaint	practitioner	The source for recording a stoppage once, seen by production and maintenance alike.	The production system often knows a line stopped with a reason code while the maintenance system still shows the machine as available; connecting them makes one stoppage drive both.	15
SRC017	OEE data accuracy best practices	OxMaint	practitioner	Establishes that stoppage reasons must come from a governed list.	Downtime reasons must be standardised into tiers; when operators pick from many vague reasons, a changeover is logged four different ways.	7

ID	Title	Organisation	Type	Why included	Establishes	Cited by
SRC018	Food plant production uptime	OxMaint	practitioner	Food-specific evidence on critical machines and failure recording.	Equipment effectiveness is availability times performance times quality; food plants start with structured failure codes on every stoppage and focus on their most critical machines.	19
SRC019	ISO 22716 batch traceability	Registrar Corp	standard	Evidence that the traceability structures used in food also apply to cosmetics.	ISO 22716 requires documentation of every stage of cosmetic production from raw materials to distribution, and US cosmetics law now aligns with it on traceability.	1
SRC020	ISO 22716 and raw material management	Registrar Corp	standard	The main source on qualifying suppliers and checking what arrives.	Raw materials require supplier qualification, testing against specification, documented receipt, controlled storage, and batch tracking with deviation management.	40
SRC021	Guidance for Industry: Cosmetic Good Manufacturing Practices	US Food and Drug Administration	regulatory	The US regulator's position on cosmetics records.	Documentation should capture every aspect of manufacturing so that problems can be traced and corrected.	1
SRC022	ISO 22716 — Cosmetics GMP	SG Systems Global	standard	The only source putting outside manufacturers under the same good-practice rules as own plants.	ISO 22716 covers personnel, premises, equipment and maintenance, materials, production, quality control, contract and outsourced activities, recalls and complaints, and expects any batch to be reconstructable.	18
SRC023	Turnkey vs. Tolling — Which Makes Sense for Me?	Ryder	practitioner	Defines the two arrangements. The scope settles on turnkey.	In tolling the brand supplies its own raw materials; in turnkey manufacturing the contract manufacturer provides them.	7

ID	Title	Organisation	Type	Why included	Establishes	Cited by
SRC024	Contract Manufacturing Solutions	Ryder	practitioner	States who does what in turnkey manufacturing.	In turnkey contract manufacturing the manufacturer sources materials, makes, checks and packs the product; the client provides specifications and approvals.	9
SRC025	California Cleaning Products Right to Know Act	Davis Wright Tremaine	regulatory	The household-products regime for ingredient disclosure — a subgroup otherwise unsourced.	Household cleaning manufacturers must disclose each ingredient, its CAS number, its function and a link to its safety data sheet; some may be withheld as confidential under a generic name.	6
SRC026	Cleansing Product Information Disclosure Program: Recommended Best Management Practices	New York State Department of Environmental Conservation	regulatory	A second state regime confirming ingredient disclosure by weight for household products.	Manufacturers of household cleansing products furnish each ingredient and its content by weight; confidential business information may be withheld.	3
SRC027	FSMA: legal ramifications of cGMPs are important to pet food safety	Michigan State University Extension	regulatory	The regulatory basis for pet food manufacturing — a subgroup otherwise unsourced.	Facilities making animal food, including pet food, must comply with good manufacturing practice under 21 CFR Part 507.	1
SRC028	Managing pet food ingredients	Food Engineering	practitioner	Evidence that pet food carries the same ingredient and control structures as human food.	Pet food uses more than 500 ingredients; manufacturers identify hazards and put preventive controls such as heat treatment in place; the FDA regulates both finished pet food and its ingredients.	1
SRC029	Good pet food manufacturing practice	UK Pet Food	practitioner	Industry-body evidence for supplier approval and batch records in pet food.	Responsible pet food makers run supplier approval, precise formulation, and batch records of finished product and where it went.	4

ID	Title	Organisation	Type	Why included	Establishes	Cited by
SRC030	The GTIN management handbook	GS1 UK	standard	The external rule for when a change makes a new product rather than a new version.	A change to declared formulation — such as adding nuts, which introduces an allergen — net content, a certification mark or pack quantity requires a new product number at every packing level.	26
SRC031	GTIN allocation rules	GS1 New Zealand	standard	Confirms the same rule and defines formulation.	Formulation is the list of ingredients or components used to create a trade item; any change to the declared net content requires a new product number.	8
SRC032	21 CFR Part 11	SG Systems Global	regulatory	The regulatory basis for signatures and for never altering a recorded entry.	Part 11 sets when electronic records and signatures are acceptable — identity, authority, auditability and retention — across food, cosmetics and other regulated manufacturing.	10
SRC033	Electronic Batch Record	CASRAI	reference-model	Defines what a batch record must hold.	A batch record documents each step — by whom or by which equipment, when, with what materials and quantities, and with what results — including checks by a second qualified person.	27
SRC034	CPG Sec. 555.250 Statement of Policy for Labeling and Preventing Cross-contact of Common Food Allergens	US Food and Drug Administration	regulatory	The regulator's statement that the order products run in is a safety matter.	Allergens must be declared; cross-contact arises from rework, carry-over on shared equipment and production sequencing.	9
SRC035	FDA Releases Draft Compliance Policy Guide for Major Food Allergen Labeling and Cross-contact	Hogan Lovells	regulatory	Establishes that an allergen is declared by its source, including when it arrives inside another ingredient.	Nine major food allergens must be declared by the food they come from — whey must be declared as milk.	6

ID	Title	Organisation	Type	Why included	Establishes	Cited by
SRC036	FDA Issues Two Guidance Documents on Food Allergen Labeling Requirements	Hyman, Phelps & McNamara (FDA Law Blog)	regulatory	The only source stating that hazard declaration rules differ by subgroup.	Food allergen labelling rules do not apply to pet foods, animal feeds, cosmetics, drugs or household cleaning products.	1
SRC037	GS1 2D code printing	Linx	standard	Evidence that individual serial numbers are becoming readable on consumer packs.	A GS1 2D code carries the product number, batch, production date, expiry date and serial number; retailers are to accept 2D codes by the end of 2027.	8
SRC038	GS1 Sunrise 2027	QRStuff	practitioner	Establishes that individual identification is expected commercially rather than required by law.	Sunrise 2027 is not a legal requirement, but major retailers are aligning to it, with packs carrying both codes during the transition.	1
SRC039	Guide to batch tracking	Cleverence	practitioner	The source for what is captured when material arrives and where it is kept.	At receipt, capture supplier lot, quantity, expiry and certificate reference; record exact storage locations; issue lot-tracked materials to production orders.	26
SRC040	Traceability — End-to-End Lot Genealogy	SG Systems Global	practitioner	Establishes that stock carries a status such as quarantined or released, separate from its identity.	Traceability links identity (product number, lot, serial), status (quarantined, released) and events (receipt, consume, transform, pack, ship, return).	8

Appendix B — DDA Domain Definitions

Domain	DDA	Definition
People	DMD000000001	People (incl Family Units). The People Domain contains information about individuals.
Legal Entities	DMD000000012	Legal Entities (incl. Relationships, Companies, Funds, Charities, Governments, Agencies etc.). The LegalEntities Domain defines the different entities involved in business transactions, services or other processes in any capacity.
Documents	DMD000000016	Documents (including Contracts, Agreements, Amendments, etc). The Documents Domain contains records that provide documentary evidence or support for business processes including Legal, Transactional, Regulatory, Identification and Commercial.
Events	DMD000000020	Events describe meaningful events in the lifecycle of various data assets including Mergers, Births, Deaths, Stock-Splits etc.
Transactions	DMD000000022	Transactions are records of an individual trade/transaction which represents a transfer of ownership or investment usually in return for capital.

Appendix C — Standards

The standards as read from DDA on 22 September 2026. Each is cited by its record rather than copied, so the text a reader sees is the text the platform holds.

Standard	Code	Name	Category
DMS000000141	DataSharedEnterpriseAsset	Data is a shared, valuable enterprise asset	Data Principles
DMS000000142	EveryoneResponsibleForData	Everyone has a role in good data management	Data Principles
DMS000000143	PolicyOwnedByCDO	The Chief Data Officer owns and approves the Data Governance Policy	Policy Governance
DMS000000144	PolicyAlignedWithOtherPolicies	Data Governance Policy aligns with and defers to more specific policies	Policy Governance
DMS000000145	PolicyAppliesToCriticalData	Data Governance Policy applies to all critical business data	Policy Scope
DMS000000146	GovernanceCommitteeEstablished	A Data Governance Committee must be established	Governance Structure
DMS000000147	SubjectBasedDataDomains	Data domains must be defined based on the subject of the data	Data Domains
DMS000000148	CommonDataRolesDefined	Standard data governance roles must be defined	Roles and Responsibilities
DMS000000149	SingleDataModelPerDomain	Each data domain must have a single authoritative data model	Data Modeling
DMS000000150	AuthoritativeRepositoryDefined	Each data domain must have an Authoritative Data Repository	Authoritative Sources
DMS000000151	PrimaryIdentifiersEstablished	Primary record identifiers must be established by domain	Data Identification
DMS000000152	CriticalDataElementsIdentified	Critical Data Elements must be identified and governed	Data Quality
DMS000000153	IssuesManagementProgram	A Data Issues for Management Resolution program must be established	Issue Management
DMS000000154	PolicyChangeControlled	Changes to the Data Governance Policy must be formally controlled	Policy Lifecycle
DMS000000156	DataDomainDefinition	Definition of Data Domain	Glossary
DMS000000157	DataModelDefinition	Definition of Data Model	Glossary
DMS000000158	DataConceptDefinition	Definition of Data Concept	Glossary
DMS000000159	DataAttributeDefinition	Definition of Data Attribute	Glossary
DMS000000160	DataProducerDefinition	Definition of Data Producer	Glossary

Standard	Code	Name	Category
DMS000000161	DataConsumerDefinition	Definition of Data Consumer	Glossary
DMS000000162	DataCustodianDefinition	Definition of Data Custodian	Glossary
DMS000000163	OperationalDataStore Definition	Definition of Operational Data Store (ODS)	Glossary
DMS000000164	MasterDataManagement Definition	Definition of Master Data Management (MDM)	Glossary
DMS000000165	DataWarehouseDefinition	Definition of Data Warehouse	Glossary
DMS000000166	DataLakeDefinition	Definition of Data Lake	Glossary
DMS000000167	CriticalDataElementDefinition	Definition of Critical Data Element (CDE)	Glossary
DMS000000168	DataLineageStandard	Data lineage must be defined and maintained for all critical data	Data Lineage
DMS000000169	DataModelDesignPrinciples	Data Model Design Principles	APPENDIX

The practice's own method rules are held in `boilerplate.json` , which this run does not carry.

Appendix D — Subject Domain Proposals

Products · PRD

Products are the things an organisation buys, makes or sells, at every level of processing and packing — ingredients, packaging components, part-made goods, finished packs and cases — together with what each is defined to be and what it is made from. Includes the batches in which they are made or received. Excludes the individual serial-numbered units made to a product definition, which are Instances, and the financial instruments an organisation issues or trades, which are Instruments.

Admission gate	Answer
Exists independently	A product definition exists whether or not this organisation makes anything; a supplier's finished product is this organisation's ingredient.
Survives its end	It survives discontinuation as the record of what was sold, what it contained and who it was sold to.
Named unprompted	Technical, quality, procurement and supply chain roles name products every day without being led.

Test	Answer	Evidence
DT1	Products are things the business deals in. Making, testing and releasing are activities that act on them and are held as occurrences in Events.	SRC005 · SRC008
DT2	A product can be described without naming a second party — what it is made from, what it must be, how long it lasts. Supplier and customer are roles of the parties to it, not of it.	SRC031 · SRC020
DT3	A product is not a Legal Entity, a Person, a Document, a Transaction or an Event. Nothing in the current register has products as members.	SRC005
DT4	A product definition exists independently of this organisation: it is bought and sold between organisations, and a supplier's finished product is this organisation's input.	SRC023 · SRC024
DT5	After discontinuation the definition, its batches and their lineage remain producible, and are required to be for two years after the record is made.	SRC011 · SRC009
DT6	Technical, quality and procurement roles name products, ingredients and packs unprompted.	SRC020 · SRC029
DT7	Products carry their own identifier scheme — the trade item number, allocated by the organisation under GS1 rules, with internal material numbers as secondary keys.	SRC030 · SRC031
DT8	Products carry obligations no existing domain carries: declared formulation, ingredient disclosure, hazard declaration, shelf life and lot-level traceability, each with its own regulator and steward.	SRC034 · SRC025 · SRC009
DT9	Commercial, trade and marketing models all reference the same product records; a product is referenced far beyond manufacturing.	SRC008 · SRC030
DT10	No current domain's definition covers a thing that is bought, made and sold. Instruments covers financial instruments, which products are not.	SRC005

Domains considered instead

Domain	Why it did not fit
—	Instruments covers financial instruments — what an organisation issues or trades — not physical goods it makes.
DMD000000022	Transactions covers exchanges of goods, not the goods themselves; a product exists between transactions.
DMD000000016	Documents covers documentary evidence about things, not the things.

Instances · UNT

Instances are individually identified physical things — a serial-numbered pack, a machine, a vessel, a tool — each made to a product definition and traceable as itself for its whole life. Includes what is fitted into each and what condition it is in. Excludes the product definition it is made to, which is a Product, and the batch it belongs to where it is not individually identified, which is a concept of Products.

Admission gate	Answer
Exists independently	A machine or a serialised pack exists independently of the organisation that made or owns it, and is bought, sold and moved between organisations.
Survives its end	Its record survives disposal as the history of what it was fitted to, how it failed and which recalls reached it.
Named unprompted	Maintenance leads, service engineers and recall coordinators name individual machines and units every day.

Test	Answer	Evidence
DT1	An individual unit is a thing, not an activity. Maintaining and calibrating are activities held as occurrences in Events.	SRC018 · SRC016
DT2	A machine can be described without naming a second party — its serial, its hours run, how it fails.	SRC018
DT3	Not every instance is a product definition: a product has no location, condition or service life, and one definition has many units.	SRC007
DT4	A serialised pack and a machine both exist independently of this organisation; the machine was made and identified by its own manufacturer.	SRC037 · SRC018
DT5	After disposal, the unit's fittings, failures and the recalls that reached it remain answerable.	SRC018 · SRC007
DT6	Maintenance and quality roles name individual machines and serialised units unprompted.	SRC018 · SRC016
DT7	Instances carry their own identifier scheme — serial number, asset number, or the serialised trade item number carried in a 2D code.	SRC037
DT8	An individual unit carries obligations no other domain carries: calibration, service life, failure history and unit-level recall.	SRC018 · SRC016
DT9	Maintenance, finance and safety processes all reference the same machine records.	SRC016 · SRC018
DT10	No current domain covers an individually identified physical thing.	SRC007

Domains considered instead

Domain	Why it did not fit
—	Instruments covers financial instruments, not physical units.
—	Products was considered and rejected: a product is a definition, an instance is a physical thing with a place, a condition and a service life.

Equipment · EQP

Equipment covers the places in a plant where work is performed — site, area, work centre, work unit and equipment module — what each is able to do, and the rules that constrain what may run there. Includes the plant structure as ISA-95 defines it. Excludes the machines that stand in those places, which are Instances, and the geographic places goods are held at, which are Locations.

Admission gate	Answer
Exists independently	A production line exists independently of the machine standing in it and of the organisation operating it; lines are bought, sold and relocated with a site.
Survives its end	It survives the replacement of every machine in it, and a past batch is still answerable against the place it ran at.
Named unprompted	Plant managers and production planners name lines, cells and units every day.

Test	Answer	Evidence
DT1	A place in a plant is a thing, not an activity. Running and changing over are occurrences held in Events.	SRC001 · SRC004
DT2	A line can be described without naming a second party — what it can do, how fast, what it sits within.	SRC001
DT3	A place in a plant is not a Location: a location is a geographic place goods are held at, a work centre is a position in a production structure. One site holds many work centres.	SRC001
DT4	ISA-95 defines the equipment hierarchy as a role-based structure that exists independently of the machines filling it.	SRC001 · SRC004
DT5	After decommissioning, what ran at the place and what stood in it remain answerable.	SRC016 · SRC001
DT6	Plant managers name lines, cells and filling stations unprompted.	SRC018 · SRC016
DT7	Places in a plant carry their own identifier scheme, used by the control and execution systems.	SRC001
DT8	A place carries obligations no other domain carries: qualification to run a product, sequencing constraints, capacity and changeover.	SRC034 · SRC017
DT9	Production, maintenance, planning and quality all reference the same plant structure.	SRC016 · SRC004
DT10	No current domain covers the structure of a plant.	SRC001

Domains considered instead

Domain	Why it did not fit
—	Instruments is financial and unrelated.
—	Locations was considered and rejected on DT3: a site is a geographic place; a work centre is a position within a production structure, and the two have different obligations.

Processes · PRC

Processes are the defined steps of work by which products are made — mixing, pasteurising, filling, labelling — the order they run in, the settings they run at, and the recipes that state them at each level of specificity from formulation to the instructions issued for one batch. Includes what each step requires of equipment, people and materials. Excludes what a product is made from, which is Products, and the record of a step actually being performed, which is an Event.

Admission gate	Answer
Exists independently	A step of work is defined once, independently of any product using it, and of the organisation performing it; ISA-88 defines it as portable across equipment and sites.
Survives its end	It survives the discontinuation of every product that used it, and a past batch is still judged against the recipe it ran to.
Named unprompted	Technical and plant roles name steps — pasteurising, filling, capping — every day.

Test	Answer	Evidence
DT1	A step is a defined thing that exists whether or not it is running; its running is an occurrence held in Events. The distinction is ISA-88's own.	SRC002 · SRC003
DT2	A step can be described without naming a second party — what it does, what it needs, what it is set to.	SRC002
DT3	A step is not a product, a place or an event. One step is used in making many products at many places.	SRC004 · SRC003
DT4	ISA-88 defines recipes and procedures as portable between equipment and organisations, separate from the plant that runs them.	SRC002 · SRC003
DT5	After a product is discontinued the recipe it was made to remains, and is required to judge batches already made.	SRC003 · SRC033
DT6	Technical and plant roles name steps and recipes unprompted.	SRC004 · SRC003
DT7	Recipes and steps carry their own identifier scheme in recipe management and execution systems.	SRC003
DT8	A recipe carries obligations no other domain carries: approval, version control, parameter limits and the instructions issued for one batch.	SRC003 · SRC033
DT9	Planning, production, quality and regulatory processes all reference the same recipes.	SRC003 · SRC021
DT10	No current domain covers a defined method of work.	SRC002

Domains considered instead

Domain	Why it did not fit
—	Products was considered and rejected: the recipe says what goes in, the method says how — one step serves many products, so it cannot hang off any one of them.
DMD000000020	Events was considered and rejected: an event is a step being performed, which is not the step itself.

Locations · LOC

Locations are identifiable geographic places at which things are held or handled — a site, a plant, a warehouse, a storage position within one — the conditions each must keep, and the identifiers each is known by. Includes places operated by other organisations where goods are received from or delivered to. Excludes the production structure within a site, which is Equipment.

Admission gate	Answer
Exists independently	A site or warehouse exists independently of this organisation; suppliers', customers' and regulators' places are all locations and each is identified by its own authority.
Survives its end	It survives closure as the place that past movements and past production refer to.
Named unprompted	Supply chain, quality and regulatory roles name sites, plants and warehouses every day.

Test	Answer	Evidence
DT1	A place is a thing, not an activity. Storing and moving are occurrences held in Events and Transactions.	SRC039 · SRC040
DT2	A site can be described without naming a second party — where it is, what conditions it keeps.	SRC020 · SRC039
DT3	A place is not a Legal Entity: one organisation operates many places, and a place may change operator.	SRC011
DT4	Places are identified by outside authorities — a global location number, a regulator's facility registration — independently of this organisation.	SRC011
DT5	After a site closes, the movements and production that refer to it remain answerable, and lot codes carry their source place for two years.	SRC011 · SRC009
DT6	Supply chain and quality roles name sites and warehouses unprompted.	SRC039 · SRC029
DT7	Places carry their own identifier scheme — global location number, facility registration number, internal site code.	SRC011
DT8	A place carries obligations no other domain carries: storage conditions, site certification, and identification as a lot code source.	SRC020 · SRC011
DT9	Manufacturing, distribution, commercial and regulatory processes all reference the same places.	SRC011 · SRC040
DT10	No current domain covers a geographic place.	SRC011

Domains considered instead

Domain	Why it did not fit
DMD000000012	LegalEntities was considered and rejected on DT3: an organisation is not a place, and one organisation operates many.
—	Equipment was considered and rejected: the plant structure within a site is a different subject with different obligations.

Candidates that are not domains

Candidate	Fails	Why	Its subjects sit in
Quality	DT1	Quality is what the organisation does to products, not a thing it deals in. Specifications belong to the product, tests and findings are occurrences, and the authority to decide is a role.	Products
Maintenance	DT1	Maintenance is an activity. The machine is an Instance, the place is an Equipment position, and the work is an occurrence.	Instances
Production	DT1	Production is an activity. What is made is a Product, where it is made is an Equipment position, and the making is an occurrence.	Events
Inventory	DT1	Inventory is a position, not a subject: a quantity of a product, in a condition, at a place, at a time.	Transactions
Suppliers	DT2	A supplier is what an organisation is to a buyer. It is a role, established by the relationship that approves it to supply a material.	LegalEntities
Materials	DT3	Every material is a product — bought rather than sold, but the same shape. Separating them would break the chain from a finished case to a raw ingredient.	Products
Batches	DT3	Every batch is a batch of a product and has no existence apart from it.	Products
Recipes	DT3	A recipe is a method stated at a level of specificity, held in Processes with the steps it arranges.	Processes

Appendix E — Business Test Cases

One row per requirement a case reaches, so every requirement shows its own evidence.

REQ ID	Case ID	Question	Kind	As at	Walked through	Result	Note
REQ001	SCN013	How much of this material is available, on hold, or rejected at each site right now?	designed	as-at	ATR089 → SBJ012 → SBJ048 → ATR017 → ATR090	Pass	—
REQ002	SCN001	What goes into this product, and how much of each input ends up in one finished unit?	designed	as-at	REL001 → SBJ012 → SBJ015 → ATR001 → ATR002	Pass	—
REQ003	SCN001	What goes into this product, and how much of each input ends up in one finished unit?	designed	as-at	REL001 → SBJ012 → SBJ015 → ATR001 → ATR002	Pass	—
REQ003	SCN002	Which finished products contain this ingredient, by any route?	designed	current	REL001	Pass	—
REQ004	SCN003	Which products use this intermediate, and are any using a variant of it that should be its own product?	designed	current	REL001	Pass	—
REQ005	SCN001	What goes into this product, and how much of each input ends up in one finished unit?	designed	as-at	REL001 → SBJ012 → SBJ015 → ATR001 → ATR002	Pass	—
REQ005	SCN002	Which finished products contain this ingredient, by any route?	designed	current	REL001	Pass	—
REQ006	SCN004	How much more or less of each input did this batch use than the recipe specified?	designed	as-at	ATR001 → SBJ051 → SBJ052 → REL029 → ATR011 → ATR012	Pass	—

REQ ID	Case ID	Question	Kind	As at	Walked through	Result	Note
REQ007	SCN005	Which batches used an alternative input, and for which ingredient?	designed	as-at	ATR003 → SBJ022 → SBJ052 → ATR005 → ATR042	Pass	—
REQ008	SCN006	Was this supplier approved for this material on the day it was received?	designed	as-at	REL003 → SBJ005 → SBJ007 → REL017 → ATR007 → ATR080	Pass	—
REQ009	SCN007	Which of this batch's stated properties were tested here, and which were taken on the supplier's word?	designed	current	SBJ010 → SBJ011 → REL037 → ATR014 → ATR083	Pass	—
REQ010	SCN008	Which products depend on a bought-in ingredient whose contents we rely on the supplier to declare?	designed	current	SBJ011 → ATR018 → ATR036	Pass	—
REQ010	SCN037	What is inside the chocolate coating we buy in finished?	awkward	as-at	ATR036 → SBJ011 → ATR018	Accepted limit	The chain of what goes into what stops at anything bought in. Only what the supplier has declared is known here, and that is recorded with who declared it and whether it was checked.
REQ010	SCN038	A supplier recalls a sugar batch used inside a coating they sold us. Which of our batches are affected?	awkward	as-at	ATR012 → SBJ016 → SBJ063 → REL029 → REL030 → ATR029	Accepted limit	Our record reaches the coating batch we received. Whether that batch contains the recalled sugar is in the supplier's records, so the twenty-four-hour answer depends on them.
REQ011	SCN009	What allergens, hazards and certifications does this product carry, and from which inputs?	designed	as-at	REL004 → SBJ018 → SBJ020 → ATR034 → ATR035	Pass	—

REQ ID	Case ID	Question	Kind	As at	Walked through	Result	Note
REQ012	SCN009	What allergens, hazards and certifications does this product carry, and from which inputs?	designed	as-at	REL004 → SBJ018 → SBJ020 → ATR034 → ATR035	Pass	—
REQ013	SCN010	What must be disclosed for this household product, and what is withheld?	designed	current	SBJ019 → ATR037 → ATR038	Pass	—
REQ014	SCN009	What allergens, hazards and certifications does this product carry, and from which inputs?	designed	as-at	REL004 → SBJ018 → SBJ020 → ATR034 → ATR035	Pass	—
REQ015	SCN011	What is the latest date this finished batch can be used, given what went into it?	designed	current	SBJ021 → ATR031 → ATR041	Pass	—
REQ016	SCN012	Which batches came from this one, and which batches made up this combined one?	designed	current	SBJ016 → SBJ017 → ATR030 → ATR032	Pass	—
REQ017	SCN013	How much of this material is available, on hold, or rejected at each site right now?	designed	as-at	ATR089 → SBJ012 → SBJ048 → ATR017 → ATR090	Pass	—
REQ018	SCN013	How much of this material is available, on hold, or rejected at each site right now?	designed	as-at	ATR089 → SBJ012 → SBJ048 → ATR017 → ATR090	Pass	—
REQ019	SCN014	Which pack sizes of this product exist, and did this change create a new product or a new version?	designed	as-at	REL002 → SBJ013 → SBJ023 → ATR019 → ATR043	Pass	—
REQ020	SCN014	Which pack sizes of this product exist, and did this change create a new product or a new version?	designed	as-at	REL002 → SBJ013 → SBJ023 → ATR019 → ATR043	Pass	—

REQ ID	Case ID	Question	Kind	As at	Walked through	Result	Note
REQ021	SCN015	What recipe was this batch made to, and what was it meant to contain at the time?	designed	as-at	ATR004 → SBJ015 → ATR026 → ATR027	Pass	—
REQ022	SCN004	How much more or less of each input did this batch use than the recipe specified?	designed	as-at	ATR001 → SBJ051 → SBJ052 → REL029 → ATR011 → ATR012	Pass	—
REQ022	SCN015	What recipe was this batch made to, and what was it meant to contain at the time?	designed	as-at	ATR004 → SBJ015 → ATR026 → ATR027	Pass	—
REQ023	SCN016	What exactly is inside this individual unit, and which batch did each part come from?	designed	as-at	REL014 → SBJ026 → SBJ027 → REL013 → REL015 → ATR006	Pass	—
REQ024	SCN016	What exactly is inside this individual unit, and which batch did each part come from?	designed	as-at	REL014 → SBJ026 → SBJ027 → REL013 → REL015 → ATR006	Pass	—
REQ025	SCN017	What steps, equipment, qualifications and settings does making this product need on this line?	designed	current	SBJ033 → SBJ036 → REL006 → REL007 → ATR056 → ATR060	Pass	—
REQ026	SCN017	What steps, equipment, qualifications and settings does making this product need on this line?	designed	current	SBJ033 → SBJ036 → REL006 → REL007 → ATR056 → ATR060	Pass	—
REQ027	SCN017	What steps, equipment, qualifications and settings does making this product need on this line?	designed	current	SBJ033 → SBJ036 → REL006 → REL007 → ATR056 → ATR060	Pass	—

REQ ID	Case ID	Question	Kind	As at	Walked through	Result	Note
REQ028	SCN017	What steps, equipment, qualifications and settings does making this product need on this line?	designed	current	SBJ033 → SBJ036 → REL006 → REL007 → ATR056 → ATR060	Pass	—
REQ029	SCN018	Which instructions were issued for this batch, and which formulation did they come from?	designed	as-at	REL009 → SBJ041 → SBJ042 → REL027 → ATR064 → ATR065	Pass	—
REQ030	SCN018	Which instructions were issued for this batch, and which formulation did they come from?	designed	as-at	REL009 → SBJ041 → SBJ042 → REL027 → ATR064 → ATR065	Pass	—
REQ031	SCN019	What ran on this line just before this batch, and was the line cleaned in between?	designed	as-at	ATR057 → SBJ034 → SBJ054 → ATR058 → ATR097	Pass	—
REQ032	SCN019	What ran on this line just before this batch, and was the line cleaned in between?	designed	as-at	ATR057 → SBJ034 → SBJ054 → ATR058 → ATR097	Pass	—
REQ033	SCN020	Which machine was in this place on the day this batch was made?	designed	as-at	REL011 → SBJ026 → SBJ031 → REL010 → REL012 → ATR009	Pass	—
REQ034	SCN020	Which machine was in this place on the day this batch was made?	designed	as-at	REL011 → SBJ026 → SBJ031 → REL010 → REL012 → ATR009	Pass	—

REQ ID	Case ID	Question	Kind	As at	Walked through	Result	Note
REQ035	SCN021	Which machines cause the most unplanned stoppage, and how do they fail?	designed	as-at	ATR013 → SBJ028 → SBJ029 → REL031 → ATR050 → ATR051	Pass	—
REQ036	SCN021	Which machines cause the most unplanned stoppage, and how do they fail?	designed	as-at	ATR013 → SBJ028 → SBJ029 → REL031 → ATR050 → ATR051	Pass	—
REQ037	SCN022	How long did this line stop, why, and what did maintenance do about it?	designed	as-at	ATR013 → SBJ055 → SBJ056 → REL031 → ATR099 → ATR100	Pass	—
REQ038	SCN022	How long did this line stop, why, and what did maintenance do about it?	designed	as-at	ATR013 → SBJ055 → SBJ056 → REL031 → ATR099 → ATR100	Pass	—
REQ039	SCN021	Which machines cause the most unplanned stoppage, and how do they fail?	designed	as-at	ATR013 → SBJ028 → SBJ029 → REL031 → ATR050 → ATR051	Pass	—
REQ040	SCN023	Who carried out each step of this batch, and were they qualified on that day?	designed	as-at	REL020 → SBJ001 → SBJ002 → REL021 → REL026 → ATR010	Pass	—
REQ040	SCN040	Two products are recorded as running at the same moment on a line and on a unit inside it. Which was it?	awkward	as-at	REL020 → SBJ001 → SBJ002 → REL021 → REL026 → ATR010	Pass	Both runs are held with their places and times, and the place hierarchy says one is inside the other. Whether they may overlap is a value-level rule, authored against the model in DDA.

REQ ID	Case ID	Question	Kind	As at	Walked through	Result	Note
REQ040	SCN041	An operator ran a step the day after their qualification expired.	awkward	as-at	REL020 → SBJ001 → SBJ002 → REL021 → REL026 → ATR010	Pass	Recorded as it happened. The qualification's dates show it had expired, so the run is answerable rather than blocked after the event.
REQ041	SCN023	Who carried out each step of this batch, and were they qualified on that day?	designed	as-at	REL020 → SBJ001 → SBJ002 → REL021 → REL026 → ATR010	Pass	—
REQ041	SCN040	Two products are recorded as running at the same moment on a line and on a unit inside it. Which was it?	awkward	as-at	REL020 → SBJ001 → SBJ002 → REL021 → REL026 → ATR010	Pass	Both runs are held with their places and times, and the place hierarchy says one is inside the other. Whether they may overlap is a value-level rule, authored against the model in DDA.
REQ041	SCN041	An operator ran a step the day after their qualification expired.	awkward	as-at	REL020 → SBJ001 → SBJ002 → REL021 → REL026 → ATR010	Pass	Recorded as it happened. The qualification's dates show it had expired, so the run is answerable rather than blocked after the event.
REQ042	SCN024	Where has this batch been, and where is it now?	designed	as-at	ATR085 → SBJ043 → SBJ046 → REL016 → REL023 → ATR069	Pass	—
REQ044	SCN025	Did this batch meet the specification in force when it was made, and by what test?	designed	as-at	ATR023 → SBJ014 → SBJ039 → ATR024 → ATR025	Pass	—
REQ045	SCN025	Did this batch meet the specification in force when it was made, and by what test?	designed	as-at	ATR023 → SBJ014 → SBJ039 → ATR024 → ATR025	Pass	—

REQ ID	Case ID	Question	Kind	As at	Walked through	Result	Note
REQ046	SCN026	What went wrong on this batch — a procedure not followed, the product out of specification, or both?	designed	current	SBJ058 → SBJ059 → REL032 → ATR106 → ATR107	Pass	—
REQ047	SCN027	What was decided about this failed material, why, and by whom?	designed	current	SBJ060 → REL028 → REL032 → ATR010 → ATR110	Pass	—
REQ048	SCN026	What went wrong on this batch — a procedure not followed, the product out of specification, or both?	designed	current	SBJ058 → SBJ059 → REL032 → ATR106 → ATR107	Pass	—
REQ049	SCN028	Which corrective actions are open, and were any closed without checking they worked?	designed	current	SBJ009 → SBJ061 → REL034 → REL036 → ATR082 → ATR112	Pass	—
REQ050	SCN029	Who released this batch, when, and on what basis?	designed	current	SBJ062 → REL028 → ATR010 → ATR115	Pass	—
REQ051	SCN030	Every batch this ingredient batch went into, and everywhere those went — within twenty-four hours	designed	current	SBJ016 → SBJ063 → REL029 → REL030 → ATR012 → ATR029	Pass	—
REQ051	SCN038	A supplier recalls a sugar batch used inside a coating they sold us. Which of our batches are affected?	awkward	as-at	ATR012 → SBJ016 → SBJ063 → REL029 → REL030 → ATR029	Accepted limit	Our record reaches the coating batch we received. Whether that batch contains the recalled sugar is in the supplier's records, so the twenty-four-hour answer depends on them.
REQ052	SCN030	Every batch this ingredient batch went into, and everywhere those went — within twenty-four hours	designed	current	SBJ016 → SBJ063 → REL029 → REL030 → ATR012 → ATR029	Pass	—

REQ ID	Case ID	Question	Kind	As at	Walked through	Result	Note
REQ052	SCN038	A supplier recalls a sugar batch used inside a coating they sold us. Which of our batches are affected?	awkward	as-at	ATR012 → SBJ016 → SBJ063 → REL029 → REL030 → ATR029	Accepted limit	Our record reaches the coating batch we received. Whether that batch contains the recalled sugar is in the supplier's records, so the twenty-four-hour answer depends on them.
REQ053	SCN030	Every batch this ingredient batch went into, and everywhere those went — within twenty-four hours	designed	current	SBJ016 → SBJ063 → REL029 → REL030 → ATR012 → ATR029	Pass	—
REQ053	SCN038	A supplier recalls a sugar batch used inside a coating they sold us. Which of our batches are affected?	awkward	as-at	ATR012 → SBJ016 → SBJ063 → REL029 → REL030 → ATR029	Accepted limit	Our record reaches the coating batch we received. Whether that batch contains the recalled sugar is in the supplier's records, so the twenty-four-hour answer depends on them.
REQ054	SCN031	Which complaints concern this batch, and what else was made from the same inputs?	designed	current	SBJ064 → REL033 → ATR118	Pass	—
REQ055	SCN032	Who signed this record, what did it mean, and has anything been corrected since?	designed	as-at	ATR077 → SBJ004 → SBJ065 → REL035 → ATR078 → ATR079	Pass	—
REQ056	SCN032	Who signed this record, what did it mean, and has anything been corrected since?	designed	as-at	ATR077 → SBJ004 → SBJ065 → REL035 → ATR078 → ATR079	Pass	—
REQ057	SCN032	Who signed this record, what did it mean, and has anything been corrected since?	designed	as-at	ATR077 → SBJ004 → SBJ065 → REL035 → ATR078 → ATR079	Pass	—

REQ ID	Case ID	Question	Kind	As at	Walked through	Result	Note
REQ058	SCN006	Was this supplier approved for this material on the day it was received?	designed	as-at	REL003 → SBJ005 → SBJ007 → REL017 → ATR007 → ATR080	Pass	—
REQ059	SCN028	Which corrective actions are open, and were any closed without checking they worked?	designed	current	SBJ009 → SBJ061 → REL034 → REL036 → ATR082 → ATR112	Pass	—
REQ060	SCN033	Which of our finished batches were made by an outside manufacturer, to which specification, and when did they become ours?	designed	as-at	REL005 → SBJ005 → SBJ046 → REL019 → REL022 → ATR008	Pass	—
REQ061	SCN033	Which of our finished batches were made by an outside manufacturer, to which specification, and when did they become ours?	designed	as-at	REL005 → SBJ005 → SBJ046 → REL019 → REL022 → ATR008	Pass	—
REQ062	SCN033	Which of our finished batches were made by an outside manufacturer, to which specification, and when did they become ours?	designed	as-at	REL005 → SBJ005 → SBJ046 → REL019 → REL022 → ATR008	Pass	—
REQ063	SCN008	Which products depend on a bought-in ingredient whose contents we rely on the supplier to declare?	designed	current	SBJ011 → ATR018 → ATR036	Pass	—
REQ063	SCN037	What is inside the chocolate coating we buy in finished?	awkward	as-at	ATR036 → SBJ011 → ATR018	Accepted limit	The chain of what goes into what stops at anything bought in. Only what the supplier has declared is known here, and that is recorded with who declared it and whether it was checked.

REQ ID	Case ID	Question	Kind	As at	Walked through	Result	Note
REQ063	SCN038	A supplier recalls a sugar batch used inside a coating they sold us. Which of our batches are affected?	awkward	as-at	ATR012 → SBJ016 → SBJ063 → REL029 → REL030 → ATR029	Accepted limit	Our record reaches the coating batch we received. Whether that batch contains the recalled sugar is in the supplier's records, so the twenty-four-hour answer depends on them.
REQ064	SCN023	Who carried out each step of this batch, and were they qualified on that day?	designed	as-at	REL020 → SBJ001 → SBJ002 → REL021 → REL026 → ATR010	Pass	—
REQ064	SCN040	Two products are recorded as running at the same moment on a line and on a unit inside it. Which was it?	awkward	as-at	REL020 → SBJ001 → SBJ002 → REL021 → REL026 → ATR010	Pass	Both runs are held with their places and times, and the place hierarchy says one is inside the other. Whether they may overlap is a value-level rule, authored against the model in DDA.
REQ064	SCN041	An operator ran a step the day after their qualification expired.	awkward	as-at	REL020 → SBJ001 → SBJ002 → REL021 → REL026 → ATR010	Pass	Recorded as it happened. The qualification's dates show it had expired, so the run is answerable rather than blocked after the event.
REQ065	SCN024	Where has this batch been, and where is it now?	designed	as-at	ATR085 → SBJ043 → SBJ046 → REL016 → REL023 → ATR069	Pass	—
REQ066	SCN034	Was this batch ever stored outside its permitted conditions?	designed	as-at	ATR015 → SBJ045 → SBJ066 → REL038 → ATR072 → ATR073	Pass	—
REQ067	SCN035	Which place or product is this, whatever identifier the supplier or regulator used?	designed	current	SBJ006 → SBJ013 → REL018 → ATR019 → ATR020	Pass	—

REQ ID	Case ID	Question	Kind	As at	Walked through	Result	Note
REQ069	SCN035	Which place or product is this, whatever identifier the supplier or regulator used?	designed	current	SBJ006 → SBJ013 → REL018 → ATR019 → ATR020	Pass	—
REQ070	SCN036	What did this batch cost against its standard, and why?	designed	as-at	ATR044 → SBJ024 → ATR045	Pass	—
REQ070	SCN039	What did poor quality cost across the plant last year?	awkward	as-at	ATR044 → SBJ024 → ATR045	Gap	Scrap, rework, stoppages and complaints are each held, but nothing brings them together as one measure, and adding them up is a different question.

Appendix F — Relationship Attributes

Data belonging to neither party to the arrangement.

Relationship	Attribute	Definition	Type	Lookup	Nature	As at	Mandatory	Critical	Classification	From
REL001	ProductBillOfMaterialsIsSubstitutable	Whether an approved alternative may fill this place.	BOOLEAN	—	observed	as-at	no	no	—	DRQ005
REL001	ProductBillOfMaterialsPosition	The place in the recipe this input fills.	TEXT	—	observed	as-at	yes	no	—	DRQ001 · DRQ005
REL001	ProductBillOfMaterialsSpecifiedQuantity	How much of the input goes into one of the output product.	FLOAT	—	observed	as-at	yes	yes	—	DRQ001 · DRQ004
REL001	ProductBillOfMaterialsStructureVersion	The version of the recipe this line belongs to.	TEXT	—	observed	as-at	yes	yes	—	DRQ001 · DRQ015
REL001	ProductBillOfMaterialsUnitOfMeasure	The unit the specified quantity is expressed in.	LOOKUP	Universal UnitOf Measure	observed	as-at	yes	no	—	DRQ001
REL003	ProductApprovedSourcesApprovalState	Whether the approval is in force, suspended or lapsed.	TEXT	—	observed	as-at	yes	yes	—	DRQ006
REL005	ProductContractManufactureScope	Whether the outside manufacturer makes the product, packs it, or both.	TEXT	—	observed	as-at	yes	no	—	DRQ033
REL011	EquipmentInstallationsInstalledBy	Who installed the machine at the place.	USER	—	observed	as-at	no	no	INTERNAL	DRQ020
REL015	InstanceBatchCompositionQuantity	How much of the batch was incorporated into the unit.	FLOAT	—	observed	as-at	yes	no	—	DRQ016
REL028	EventPerformersCapacity	In what capacity the person performed the work — operator, approver, releaser, second checker.	TEXT	—	observed	as-at	yes	yes	—	DRQ023 · DRQ027 · DRQ029

Relationship	Attribute	Definition	Type	Lookup	Nature	As at	Mandatory	Critical	Classification	From
REL029	EventBatchInputsQuantityUsed	How much of the input batch the run actually used.	FLOAT	—	observed	as-at	yes	yes	—	DRQ004
REL030	EventBatchOutputsQuantityProduced	How much the run produced.	FLOAT	—	observed	as-at	yes	yes	—	DRQ004 · DRQ030
REL031	EventStoppageMachinesFailureMode	How the machine failed on this stoppage.	TEXT	—	observed	as-at	no	no	—	DRQ021 · DRQ022
REL037	DocumentBatchDeclarationsVerified	Whether the organisation checked what the document states.	BOOLEAN	—	observed	as-at	yes	yes	—	DRQ007
REL038	EventStorageExposuresDuration	How long the batch was exposed to conditions outside the limits.	FLOAT	—	observed	as-at	yes	no	—	DRQ034

Appendix G — Concept Attributes

Domain	Concept	Attribute	Definition	Type	Lookup	Nature	As at	Mandatory	Critical	Classification	From
Documents	SBJ011 DocumentDeclarations	Document Declarations Property	The property the document states.	TEXT	—	observed	as-at	yes	yes	—	DRQ007
Documents	SBJ011 DocumentDeclarations	Document Declarations Value	The value the document states for it.	TEXT	—	observed	as-at	yes	yes	—	DRQ007
Equipment	SBJ033 EquipmentCapabilities	Equipment Capabilities Capacity	How much the place can handle, in its own unit.	FLOAT	—	observed	as-at	yes	no	—	DRQ017
Equipment	SBJ035 EquipmentMaintenancePlans	EquipmentMaintenancePlans Trigger	What prompts the work — a date, a count, a condition.	TEXT	—	observed	as-at	yes	no	—	DRQ021
Equipment	SBJ031 EquipmentPosition	EquipmentPositionLevel	Which level of the plant structure the place sits at.	LOOKUP	EquipmentPositionLevel Type	observed	current	yes	yes	—	DRQ020
Equipment	SBJ031 EquipmentPosition	EquipmentPositionName	The name the place is known by on the floor.	TEXT	—	observed	current	yes	no	—	DRQ020
Equipment	SBJ034 EquipmentSequencingRules	Equipment Sequencing RulesReason	Why the constraint exists, and whether it is enforced or advisory.	LONGTEXT	—	observed	as-at	yes	yes	—	DRQ019
Equipment	SBJ034 EquipmentSequencingRules	Equipment Sequencing RulesRequires Clean	Whether a clean is required between the two products.	BOOLEAN	—	observed	as-at	yes	yes	—	DRQ019
Events	SBJ049 Event	EventOccurred At	When it happened.	DATETIME	—	observed	as-at	yes	yes	—	DRQ023

Domain	Concept	Attribute	Definition	Type	Lookup	Nature	As at	Mandatory	Critical	Classification	From
Events	SBJ053 Event BatchSplits	EventBatch SplitsReason	Why the batch was split or combined.	TEXT	—	observed	as-at	yes	no	—	DRQ012
Events	SBJ054 Event Changeovers	Event Changeovers Duration	How long the changeover took.	FLOAT	—	observed	as-at	yes	no	—	DRQ019
Events	SBJ054 Event Changeovers	Event Changeovers IncludedClean	Whether the line was cleaned during the changeover.	BOOLEAN	—	observed	as-at	yes	yes	—	DRQ019
Events	SBJ064 Event Complaints	Event Complaints Report	What the person outside the organisation reported.	LONGTEXT	—	observed	as-at	yes	no	CUSTOMER	DRQ031
Events	SBJ065 Event Corrections	EventCorrec-tionsReason	Why the record needed correcting.	LONGTEXT	—	observed	as-at	yes	yes	—	DRQ032
Events	SBJ061 Event Corrective Actions	EventCorrective ActionsClosed At	When the action was closed.	DATE	—	observed	as-at	no	yes	—	DRQ028
Events	SBJ061 Event Corrective Actions	EventCorrective ActionsEffect-ivenessVerified At	When the action was shown to have worked.	DATE	—	observed	as-at	no	yes	—	DRQ028
Events	SBJ061 Event Corrective Actions	EventCorrective ActionsRoot Cause	The cause found by investigation.	LONGTEXT	—	observed	as-at	yes	yes	—	DRQ028
Events	SBJ058 Event Deviations	EventDeviations Classification	How serious the departure is.	TEXT	—	observed	as-at	yes	yes	—	DRQ026
Events	SBJ058 Event Deviations	EventDeviations ContainedAt	When the affected material was held back.	DATETIME	—	observed	as-at	no	yes	—	DRQ026

Domain	Concept	Attribute	Definition	Type	Lookup	Nature	As at	Mandatory	Critical	Classification	From
Events	SBJ060 Event Dispositions	EventDispositionsDecision	What was decided about the failed material.	LOOKUP	Event Dispositions Decision Type	observed	as-at	yes	yes	—	DRQ027
Events	SBJ060 Event Dispositions	EventDispositionsJustification	Why that decision was taken.	LONGTEXT	—	observed	as-at	yes	yes	—	DRQ027
Events	SBJ056 Event Maintenance Work	EventMaintenanceWorkStoppageReference	The stoppage this work responded to, so production loss and repair describe one event.	TEXT	—	observed	as-at	no	yes	—	DRQ022
Events	SBJ052 Event Material Consumptions	EventMaterialConsumptionsYieldVariance	The difference between what the recipe specified and what was used.	FLOAT	—	derived	as-at	yes	yes	—	DRQ004
Events	SBJ059 Event NonConformances	EventNonConformancesClassification	How serious the failure is.	TEXT	—	observed	as-at	yes	yes	—	DRQ026
Events	SBJ059 Event NonConformances	EventNonConformancesContainedAt	When the affected material was held back.	DATETIME	—	observed	as-at	yes	yes	—	DRQ026
Events	SBJ051 Event Production Runs	EventProductionRunsEndedAt	When the run ended.	DATETIME	—	observed	as-at	yes	yes	—	DRQ023
Events	SBJ051 Event Production Runs	EventProductionRunsStartedAt	When the run began.	DATETIME	—	observed	as-at	yes	yes	—	DRQ023

Domain	Concept	Attribute	Definition	Type	Lookup	Nature	As at	Mandatory	Critical	Classification	From
Events	SBJ063 Event Recalls	EventRecalls Scope	What the incident covers.	LONGTEXT	—	observed	as-at	yes	yes	—	DRQ030
Events	SBJ062 Event Releases	EventReleases Basis	What the release was made on the strength of.	LONGTEXT	—	observed	as-at	yes	yes	—	DRQ029
Events	SBJ062 Event Releases	EventReleases ReleasedAt	When the output was made available.	DATETIME	—	observed	as-at	yes	yes	—	DRQ029
Events	SBJ055 Event Stoppages	EventStop-pagesEndedAt	When production resumed.	DATETIME	—	observed	as-at	yes	yes	—	DRQ022
Events	SBJ055 Event Stoppages	EventStop-pagesReason	Why production stopped.	LOOKUP	Event Stop-pages Reason Type	observed	as-at	yes	yes	—	DRQ022
Events	SBJ055 Event Stoppages	EventStop-pagesStartedAt	When production stopped.	DATETIME	—	observed	as-at	yes	yes	—	DRQ022
Events	SBJ066 Event Storage Breaches	EventStorage Breaches StartedAt	When the conditions went outside their limits.	DATETIME	—	observed	as-at	yes	yes	—	DRQ034
Events	SBJ057 Event Tests	EventTests Conformed	Whether the result met the specification in force.	BOOLEAN	—	derived	as-at	yes	yes	—	DRQ025
Events	SBJ057 Event Tests	EventTests Method	The method the test was performed by.	TEXT	—	observed	as-at	yes	yes	—	DRQ025
Events	SBJ057 Event Tests	EventTests Result	The result the test returned.	TEXT	—	observed	as-at	yes	yes	—	DRQ025
Instances	SBJ026 Instance	InstanceBuiltOn	When the unit was made.	DATE	—	observed	as-at	yes	no	—	DRQ016

Domain	Concept	Attribute	Definition	Type	Lookup	Nature	As at	Mandatory	Critical	Classification	From
Instances	SBJ028 InstanceConditions	InstanceConditionsMeasure	What was measured — hours run, cycles, wear, calibration result.	TEXT	—	observed	as-at	yes	no	—	DRQ021
Instances	SBJ028 InstanceConditions	InstanceConditionsValue	The value measured.	FLOAT	—	observed	as-at	yes	no	—	DRQ021
Instances	SBJ030 InstanceCriticalities	InstanceCriticalitiesScore	How much the machine matters, on the organisation's scale.	INTEGER	—	observed	as-at	yes	yes	—	DRQ021
Instances	SBJ029 InstanceFailureModes	InstanceFailureModesEffect	What the failure causes when it happens.	LONGTEXT	—	observed	as-at	yes	no	—	DRQ021
Instances	SBJ027 InstanceIdentifiers	InstanceIdentifiersValue	The serial or asset number itself.	TEXT	—	observed	as-at	yes	yes	—	DRQ016
Legal Entities	SBJ008 Legal EntityAssessments	LegalEntityAssessmentsScope	What the assessment covered.	LONGTEXT	—	observed	as-at	yes	no	—	DRQ006
Legal Entities	SBJ008 Legal EntityAssessments	LegalEntityAssessmentsValidUntil	When the assessment ceases to be valid.	DATE	—	observed	as-at	yes	yes	—	DRQ006
Legal Entities	SBJ009 Legal EntitySupplierRequests	LegalEntitySupplierRequestsResponse	What the supplier said in reply.	LONGTEXT	—	observed	as-at	no	no	—	DRQ028
Locations	SBJ043 Location	LocationName	The name the place is known by.	TEXT	—	observed	current	yes	no	—	DRQ024
Locations	SBJ044 LocationIdentifiers	LocationIdentifiersIssuer	Who issued the identifier.	TEXT	—	observed	as-at	yes	no	—	DRQ035

Domain	Concept	Attribute	Definition	Type	Lookup	Nature	As at	Mandatory	Critical	Classification	From
Locations	SBJ044 LocationIdentifiers	LocationIdentifiersValue	The identifier itself.	TEXT	—	observed	as-at	yes	yes	—	DRQ035
Locations	SBJ045 LocationStorageConditions	LocationStorageConditionsLowerLimit	The lowest value the condition may reach.	FLOAT	—	observed	as-at	yes	yes	—	DRQ034
Locations	SBJ045 LocationStorageConditions	LocationStorageConditionsUpperLimit	The highest value the condition may reach.	FLOAT	—	observed	as-at	yes	yes	—	DRQ034
People	SBJ002 PersonQualificationRecords	PersonQualificationRecordsAwardedOn	When the qualification was awarded.	DATE	—	observed	as-at	yes	yes	INTERNAL_PII	DRQ023
People	SBJ002 PersonQualificationRecords	PersonQualificationRecordsExpiresOn	When the qualification expires.	DATE	—	observed	as-at	yes	yes	INTERNAL_PII	DRQ023
People	SBJ003 PersonShiftAssignments	PersonShiftAssignmentsStartedAt	When the period of work began.	DATETIME	—	observed	as-at	yes	no	INTERNAL_PII	DRQ023
People	SBJ004 PersonSignatures	PersonSignaturesIsSecondCheck	Whether this signature is the second qualified check of the step.	BOOLEAN	—	observed	as-at	yes	yes	—	DRQ032
People	SBJ004 PersonSignatures	PersonSignaturesMeaning	What the signature meant — approved, reviewed, checked.	TEXT	—	observed	as-at	yes	yes	INTERNAL_PII	DRQ032
People	SBJ004 PersonSignatures	PersonSignaturesSignedAt	When the signature was applied.	DATETIME	—	observed	as-at	yes	yes	INTERNAL_PII	DRQ032
Processes	SBJ042 ProcessParameterValues	ProcessParameterValuesValue	The value the setting is run at for this product at this place.	FLOAT	—	observed	as-at	yes	yes	—	DRQ017 DRQ018

Domain	Concept	Attribute	Definition	Type	Lookup	Nature	As at	Mandatory	Critical	Classification	From
Processes	SBJ038 ProcessParameters	ProcessParametersLowerLimit	The lowest value the setting may take.	FLOAT	—	observed	as-at	yes	yes	—	DRQ017
Processes	SBJ038 ProcessParameters	ProcessParametersUpperLimit	The highest value the setting may take.	FLOAT	—	observed	as-at	yes	yes	—	DRQ017
Processes	SBJ041 ProcessRecipeLevels	ProcessRecipeLevelsIssuedOn	When this statement of the method was issued.	DATE	—	observed	as-at	yes	no	—	DRQ018
Processes	SBJ041 ProcessRecipeLevels	ProcessRecipeLevelsLevel	How specific this statement of the method is.	LOOKUP	Process Recipe Levels Type	observed	as-at	yes	yes	—	DRQ018
Processes	SBJ040 ProcessRoutings	ProcessRoutingsName	The name the method is known by.	TEXT	—	observed	as-at	yes	no	—	DRQ017
Processes	SBJ039 ProcessSamplingPlans	ProcessSamplingPlansFrequency	How often a sample is taken at this step.	TEXT	—	observed	as-at	yes	no	—	DRQ025
Processes	SBJ036 ProcessSegment	ProcessSegmentName	The name the step is known by.	TEXT	—	observed	current	yes	no	—	DRQ017
Processes	SBJ037 ProcessSegmentRequirements	ProcessSegmentRequirementsKind	What kind of thing the step needs — a capability, a qualification, a kind of material.	TEXT	—	observed	as-at	yes	no	—	DRQ017

Domain	Concept	Attribute	Definition	Type	Lookup	Nature	As at	Mandatory	Critical	Classification	From
Products	SBJ012 Product	ProductIs Purchased	Whether the product is bought in, so that what it is made from is the supplier's to declare.	BOOLEAN	—	observed	current	yes	yes	—	DRQ008
Products	SBJ012 Product	ProductKind	What the product is to this organisation.	LOOKUP	Product KindType	observed	current	yes	yes	—	DRQ001 . DRQ013
Products	SBJ012 Product	ProductName	The name the product is known by.	TEXT	—	observed	current	yes	no	—	DRQ001
Products	SBJ016 Product Batches	ProductBatches LotCode	The lot code assigned when the batch was first packed or transformed.	TEXT	—	observed	as-at	yes	yes	—	DRQ030
Products	SBJ016 Product Batches	ProductBatches MadeOn	When the batch was made or received.	DATE	—	observed	as-at	yes	yes	—	DRQ030 . DRQ012
Products	SBJ016 Product Batches	ProductBatches QuantityMade	How much the batch contains.	FLOAT	—	observed	as-at	yes	no	—	DRQ012
Products	SBJ016 Product Batches	ProductBatches UsableUntil	The latest date the batch may be used, taken from its inputs and its shelf life.	DATE	—	derived	as-at	yes	yes	—	DRQ011

Domain	Concept	Attribute	Definition	Type	Lookup	Nature	As at	Mandatory	Critical	Classification	From
Products	SBJ024 ProductCost-ings	ProductCost-ingsActualCost	What the product actually cost, built from what went into it and what was done to it.	CURRENCY	—	derived	as-at	yes	yes	—	DRQ036
Products	SBJ024 ProductCost-ings	ProductCost-ingsStandard Cost	What the product was expected to cost, by element.	CURRENCY	—	observed	as-at	yes	yes	—	DRQ036
Products	SBJ019 ProductDisclosures	ProductDisclosuresIsWithheld	Whether the identity is withheld as confidential business information.	BOOLEAN	—	observed	as-at	yes	no	—	DRQ010
Products	SBJ019 ProductDisclosures	ProductDisclosuresSubstance	The ingredient being disclosed.	TEXT	—	observed	as-at	yes	yes	—	DRQ010
Products	SBJ019 ProductDisclosures	ProductDisclosuresWeight Band	How much of it there is, as the regime requires it to be stated.	TEXT	—	observed	as-at	yes	no	—	DRQ010
Products	SBJ018 Product Hazards	ProductHazardsIsDeclaredBy Supplier	Whether the hazard is known only because a supplier declared it.	BOOLEAN	—	observed	as-at	yes	yes	—	DRQ008 DRQ009
Products	SBJ018 Product Hazards	ProductHazardsSubstance	The substance the hazard concerns, named as it must be declared.	TEXT	—	observed	as-at	yes	yes	—	DRQ009
Products	SBJ018 Product Hazards	ProductHazardsType	The kind of hazard, which decides the declaration regime.	LOOKUP	Product Hazard Type	observed	as-at	yes	yes	—	DRQ009

Domain	Concept	Attribute	Definition	Type	Lookup	Nature	As at	Mandatory	Critical	Classification	From
Products	SBJ013 ProductIdentifiers	ProductIdentifiersExpiryDate	When the identifier ceased to be used.	DATE	—	observed	as-at	no	no	—	DRQ035
Products	SBJ013 ProductIdentifiers	ProductIdentifiersStartDate	When the identifier came into use.	DATE	—	observed	as-at	yes	no	—	DRQ035
Products	SBJ013 ProductIdentifiers	ProductIdentifiersType	The scheme the identifier belongs to.	LOOKUP	Product Identifiers Type	observed	as-at	yes	no	—	DRQ014 · DRQ035
Products	SBJ013 ProductIdentifiers	ProductIdentifiersValue	The identifier itself.	TEXT	—	observed	as-at	yes	yes	—	DRQ035
Products	SBJ025 ProductIdentity Changes	ProductIdentity ChangesMade NewProduct	Whether the change made a new product rather than a new version of what goes in.	BOOLEAN	—	observed	as-at	yes	yes	—	DRQ014
Products	SBJ025 ProductIdentity Changes	ProductIdentity Changes Reason	Why the change was made, and which rule decided it.	LONGTEXT	—	observed	as-at	yes	no	—	DRQ014
Products	SBJ020 ProductOrigins	ProductOrigins Country	The country or region of origin stated.	TEXT	—	observed	as-at	yes	no	—	DRQ009
Products	SBJ021 ProductShelf Lives	ProductShelf LivesDays	How many days the product lasts under the stated conditions.	INTEGER	—	observed	as-at	yes	yes	—	DRQ011
Products	SBJ014 Product Specifications	ProductSpecificationsEffective From	The date from which this version applies.	DATE	—	observed	as-at	yes	yes	—	DRQ025

Domain	Concept	Attribute	Definition	Type	Lookup	Nature	As at	Mandatory	Critical	Classification	From
Products	SBJ014 Product Specifications	ProductSpecificationsLimit	The limit a characteristic must fall within.	TEXT	—	observed	as-at	yes	no	—	DRQ025
Products	SBJ014 Product Specifications	ProductSpecificationsVersion	Which version of the specification this is.	TEXT	—	observed	as-at	yes	yes	—	DRQ025
Products	SBJ015 ProductStructureVersions	ProductStructureVersionsEffectiveFrom	The date from which this version of the recipe applies.	DATE	—	observed	as-at	yes	yes	—	DRQ015
Products	SBJ015 ProductStructureVersions	ProductStructureVersionsScope	The site, market or period this version applies to.	TEXT	—	observed	as-at	yes	no	—	DRQ015
Products	SBJ015 ProductStructureVersions	ProductStructureVersionsVersion	Which version of the recipe this is.	TEXT	—	observed	as-at	yes	yes	—	DRQ015
Products	SBJ017 ProductSub Batches	ProductSub BatchesQuantity	How much of the batch this part holds.	FLOAT	—	observed	as-at	yes	no	—	DRQ012
Products	SBJ022 ProductSubstitutions	ProductSubstitutionsConditions	The conditions under which the alternative may be used.	LONGTEXT	—	observed	as-at	yes	no	—	DRQ005
Products	SBJ023 ProductVariant Axes	ProductVariant AxesAxis	The dimension along which the family's members differ.	TEXT	—	observed	as-at	yes	no	—	DRQ014
Transactions	SBJ046 Transaction	TransactionKind	What kind of exchange this is.	LOOKUP	TransactionKind Type	observed	as-at	yes	yes	—	DRQ024 . DRQ033

Domain	Concept	Attribute	Definition	Type	Lookup	Nature	As at	Mandatory	Critical	Classification	From
Transactions	SBJ046 Transaction	Transaction OccurredAt	When the exchange happened.	DATETIME	—	observed	as-at	yes	yes	—	DRQ024
Transactions	SBJ046 Transaction	TransactionTitle PassesOn Receipt	Whether title passes to the organisation when the goods arrive.	BOOLEAN	—	observed	as-at	yes	yes	—	DRQ033
Transactions	SBJ048 TransactionInventory Positions	Transaction InventoryPositionsCondition	Whether the stock is available, on hold or rejected.	TEXT	—	observed	as-at	yes	yes	—	DRQ013
Transactions	SBJ048 TransactionInventory Positions	Transaction InventoryPositionsCountedAt	When the quantity was established.	DATETIME	—	observed	as-at	yes	no	—	DRQ013
Transactions	SBJ048 TransactionInventory Positions	Transaction InventoryPositionsQuantity	How much is held.	FLOAT	—	observed	as-at	yes	yes	—	DRQ013
Transactions	SBJ047 TransactionLines	Transaction LinesQuantity	How much of the product the line concerns.	FLOAT	—	observed	as-at	yes	yes	—	DRQ024